

Wrapping up RNAseq + announcements

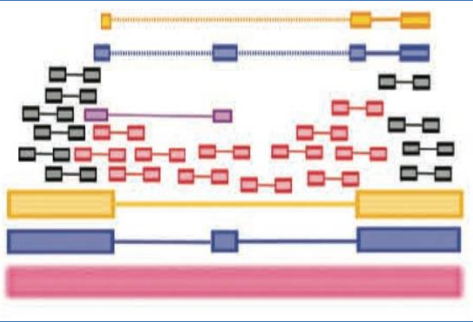
Michael Schatz

October 15, 2025

Lecture 14. Applied Comparative Genomics



RNA-seq Challenges

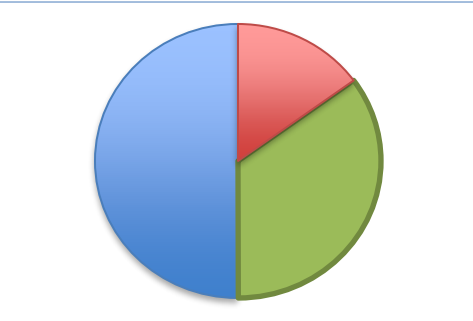


Challenge 1: Eukaryotic genes are spliced

Solution: Use a spliced aligner, and assemble isoforms

TopHat: discovering spliced junctions with RNA-Seq.

Trapnell et al (2009) *Bioinformatics*. 25:0 1105-1111

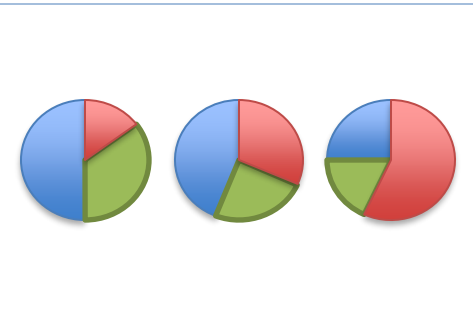


Challenge 2: Read Count != Transcript abundance

Solution: Infer underlying abundances (e.g. TPM)

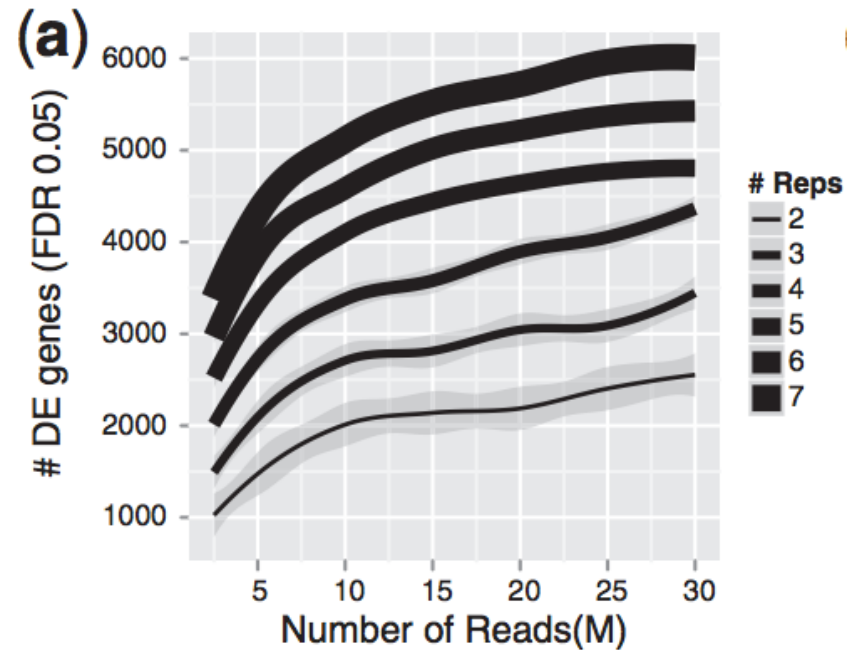
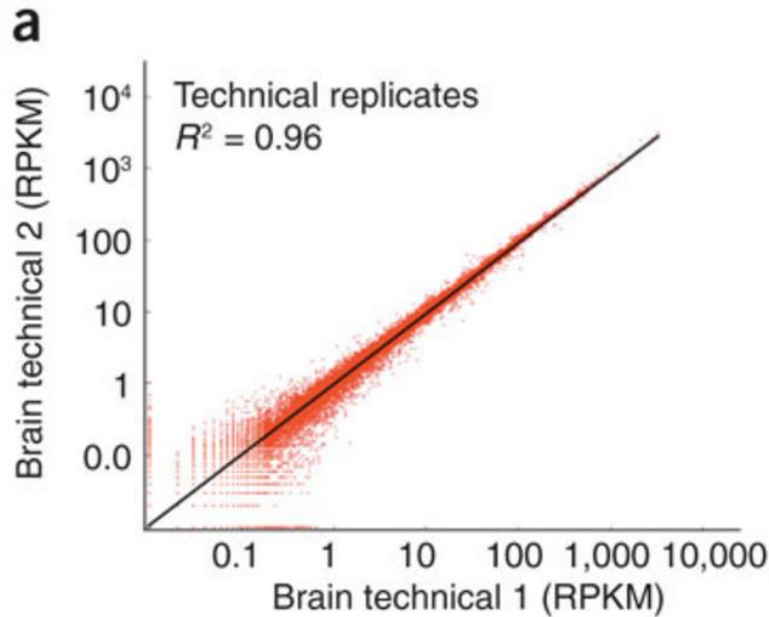
Transcript assembly and quantification by RNA-seq

Trapnell et al (2010) *Nat. Biotech.* 25(5): 511-515



Challenge 3: Transcript abundances are stochastic

How Many Replicates?



Why don't we have perfect replicates?

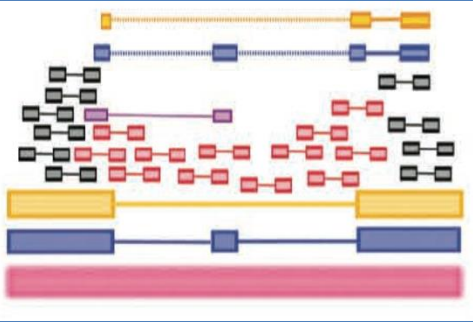
Mapping and quantifying mammalian transcriptomes by RNA-Seq

Mortazavi et al (2008) Nature Methods. 5, 62-628

RNA-seq differential expression studies: more sequence or more replication?

Liu et al (2013) Bioinformatics. doi:10.1093/bioinformatics/btt688

RNA-seq Challenges

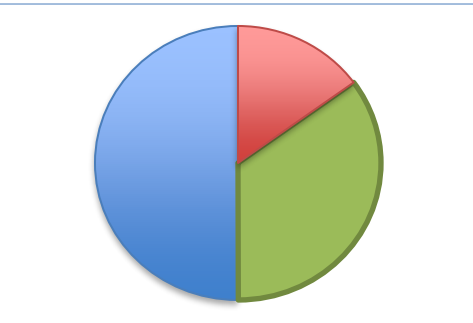


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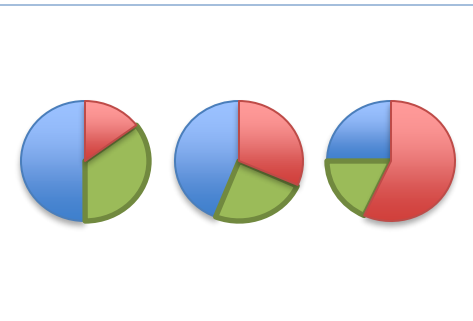


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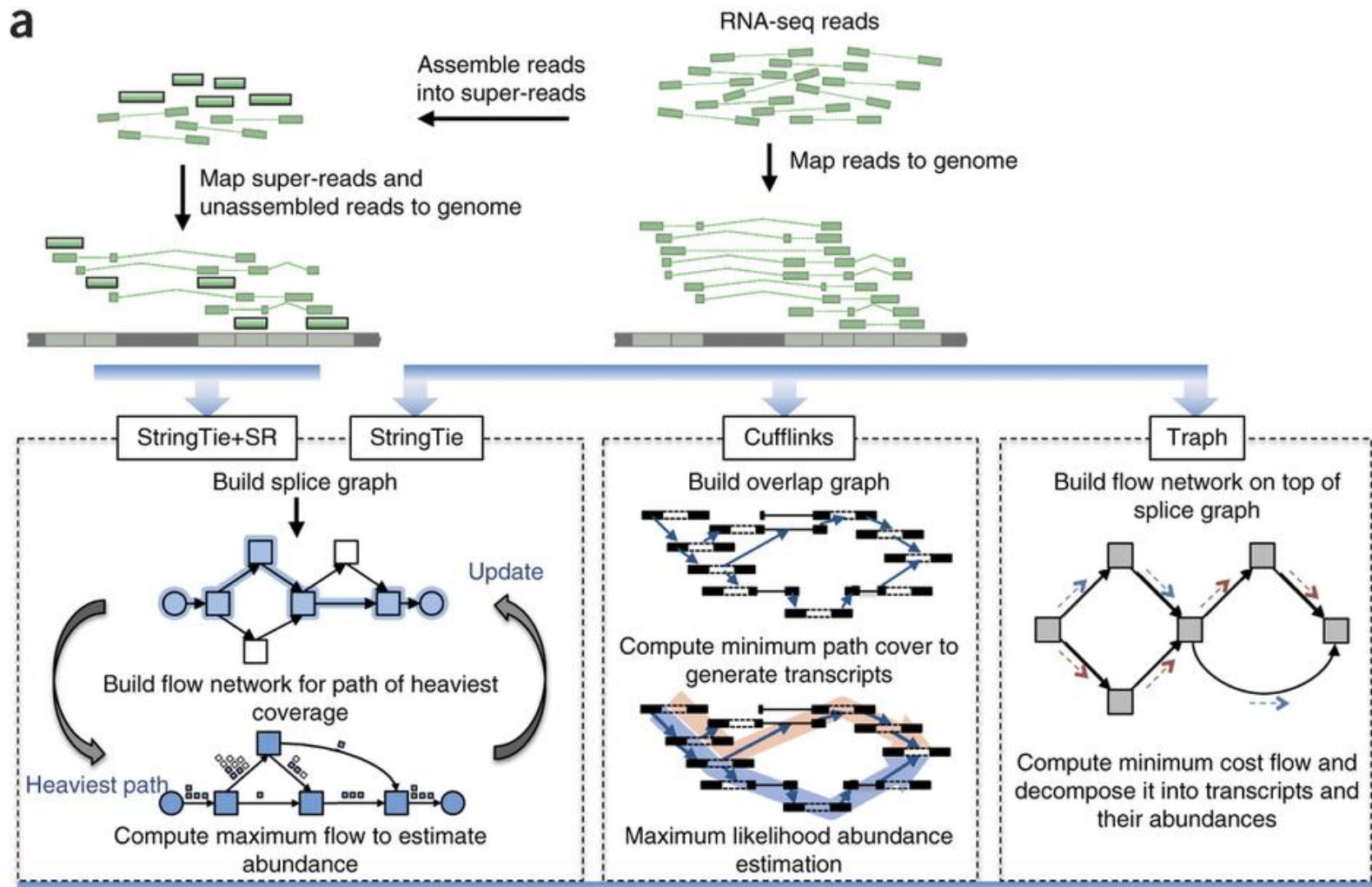
Challenge 3: Transcript abundances are stochastic

Solution: Replicates, replicates, and more replicates

RNA-seq differential expression studies: more sequence or more replication?

Liu et al (2013) *Bioinformatics*. doi:10.1093/bioinformatics/btt688

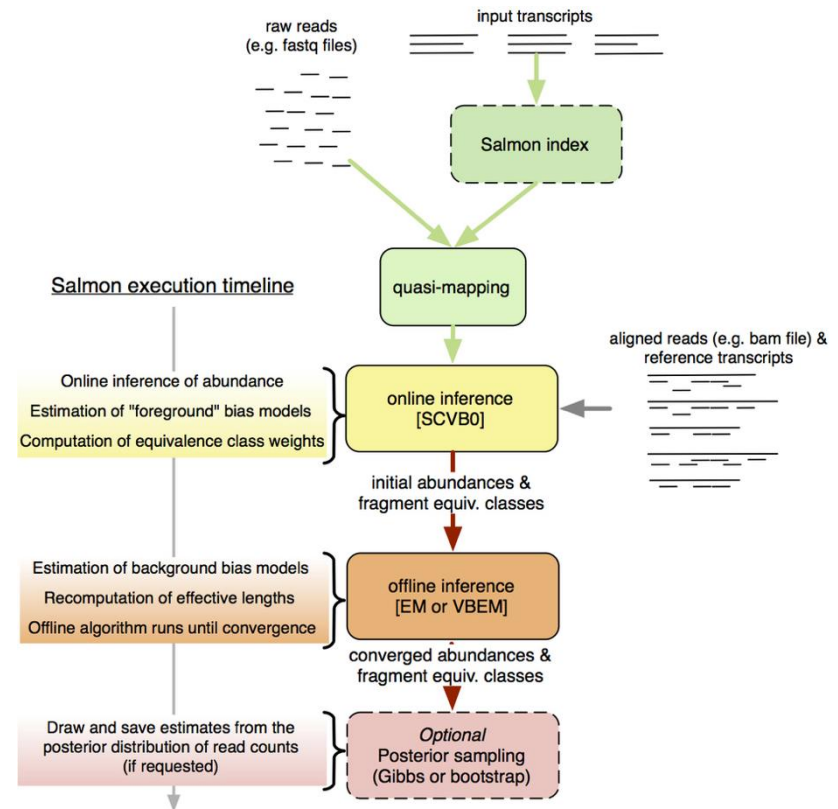
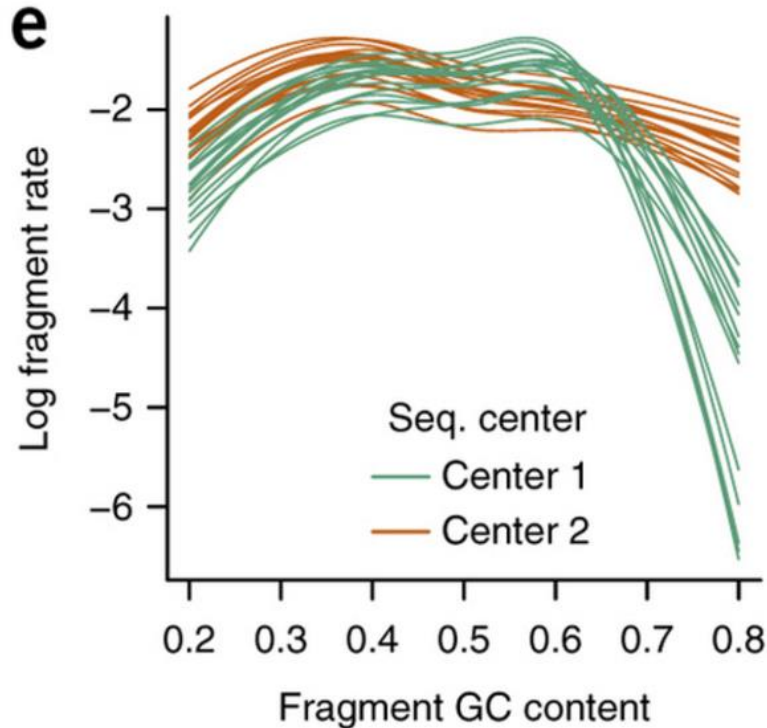
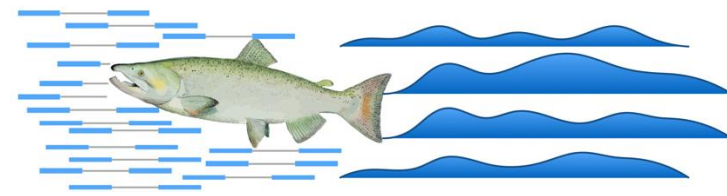
Isoform Quantification Approaches



StringTie enables improved reconstruction of a transcriptome from RNA-seq reads.

Pertea M, et al. (2015) Nature Biotechnology. doi: 10.1038/nbt.3122.

Salmon: The ultimate RNA-seq Pipeline?



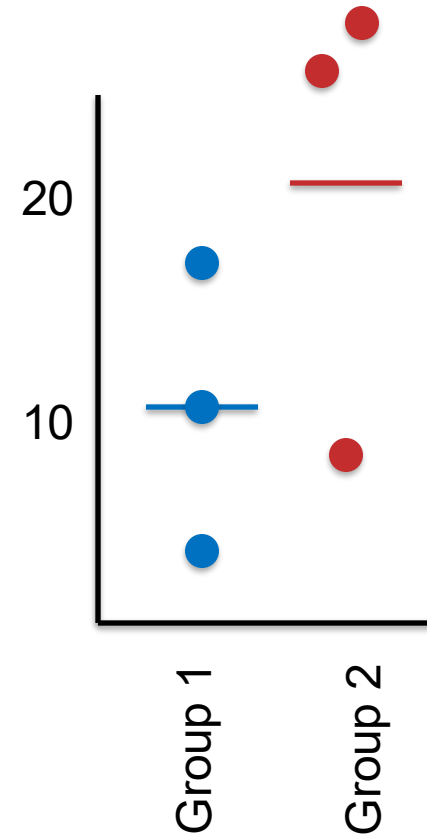
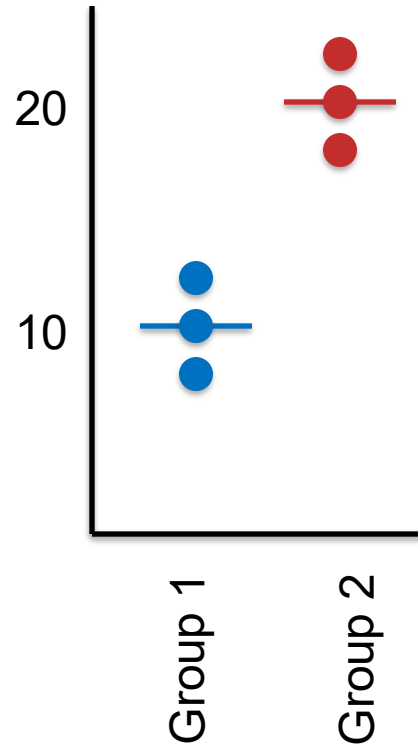
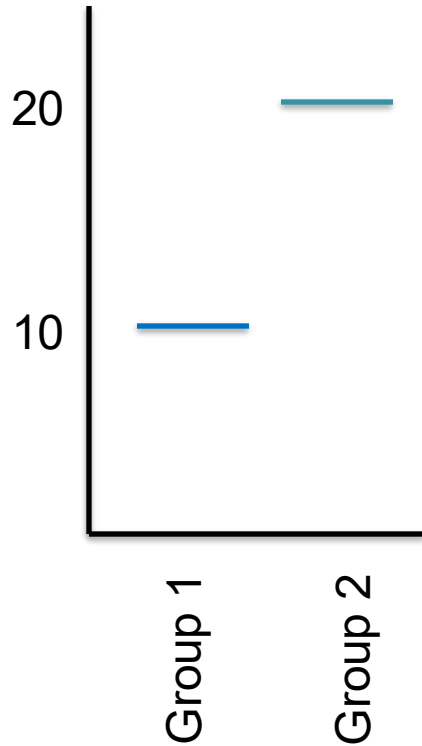
Modeling of RNA-seq fragment sequence bias reduces systematic errors in transcript abundance estimation

Love et al (2016) Nature Biotechnology 34, 1287–1291 (2016) doi:10.1038/nbt.3682

Salmon provides fast and bias-aware quantification of transcript expression

Patro et al (2017) Nature Methods (2017) doi:10.1038/nmeth.4197

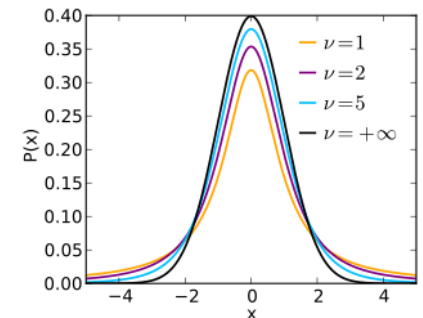
T-test



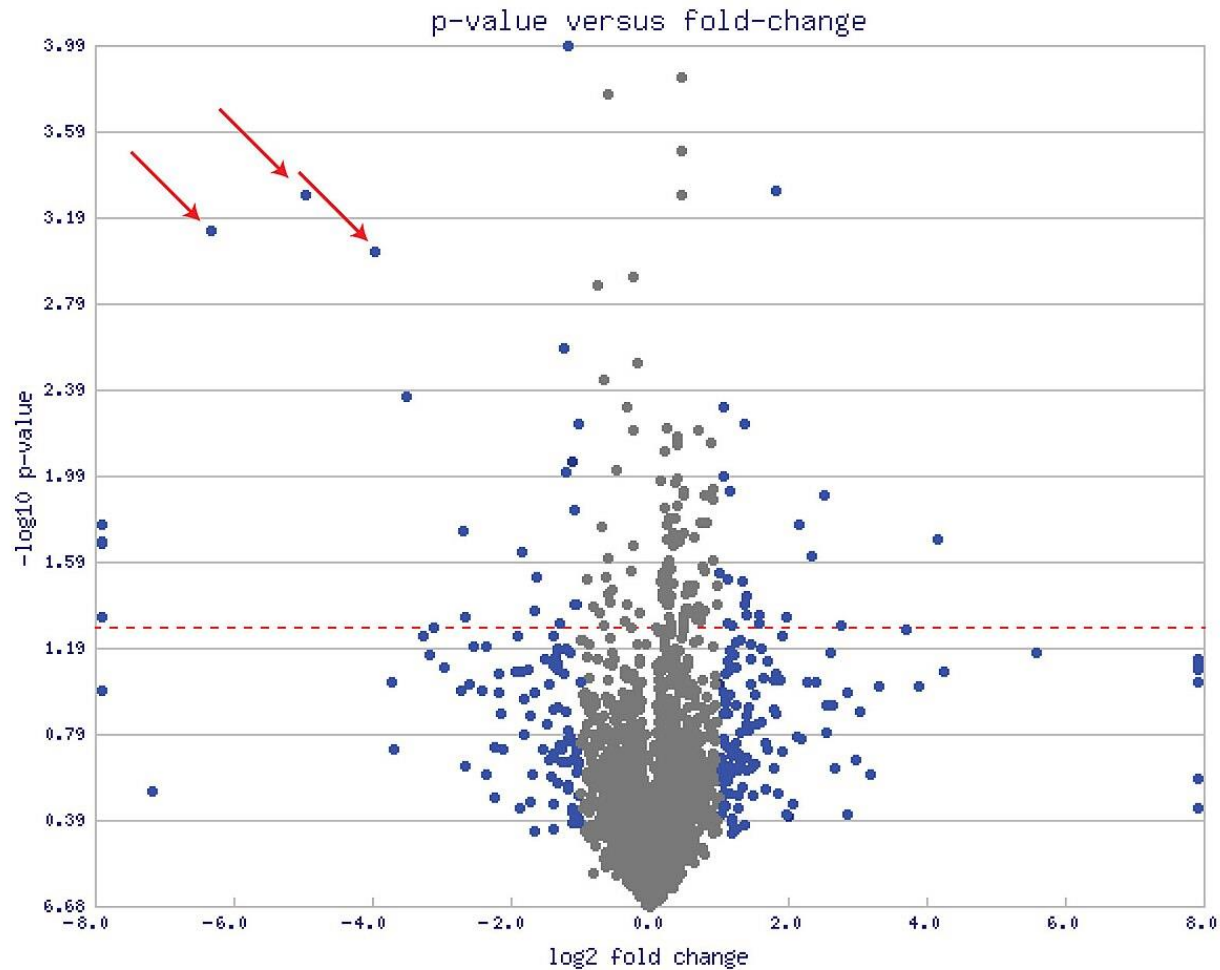
$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Where:

- $\bar{X}_1$ and $\bar{X}_2$ are the means of the two samples.
- s_1^2 and s_2^2 are the variances of the two samples.
- n_1 and n_2 are the sizes of the two samples.



Volcano Plot



https://en.wikipedia.org/wiki/Volcano_plot_%28statistics%29

Annotation Summary

- Three major approaches to annotate a genome

1. Alignment:

- Does this sequence align to any other sequences of known function?
- Great for projecting knowledge from one species to another

2. Prediction:

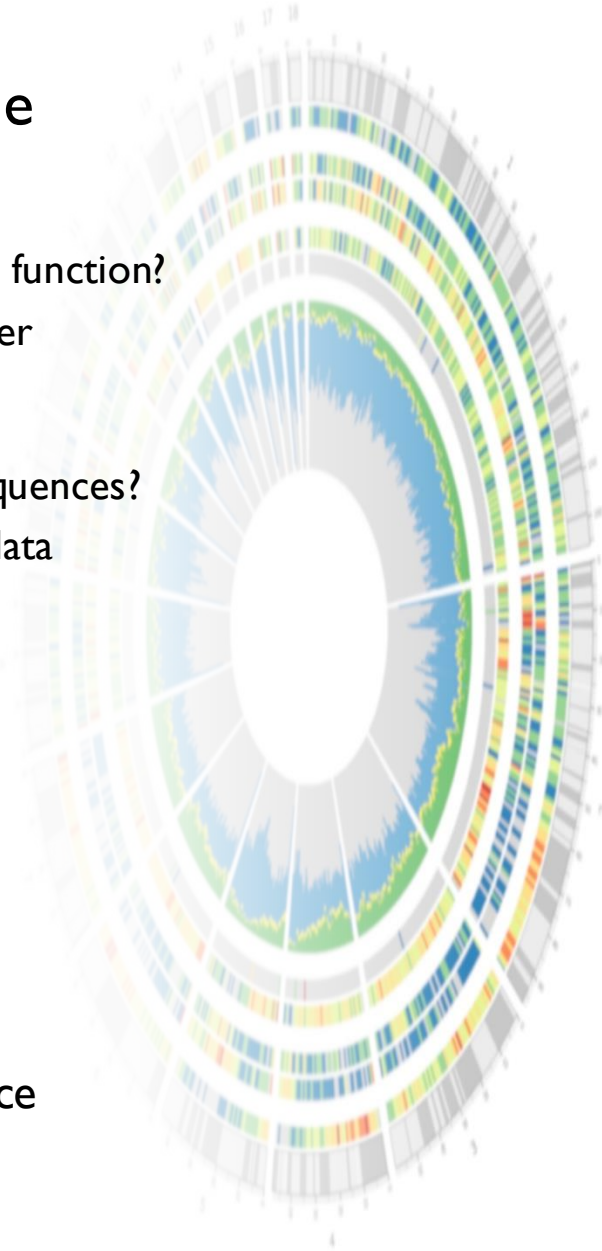
- Does this sequence statistically resemble other known sequences?
- Potentially most flexible but dependent on good training data

3. Experimental:

- Lets test to see if it is transcribed/methylated/bound/etc
- Strongest but expensive and context dependent

- Many great resources available

- Learn to love the literature and the databases
- Standard formats let you rapidly query and cross reference
- Google is your number one resource 😊



Exam grades released

DNA palindrome

- **Not a literal** palindrome
- Sequence equal to its **reverse complement**
 - Forward strand read 5' to 3' (left to right) is the same as the reverse strand read 3' to 5' (right to left)

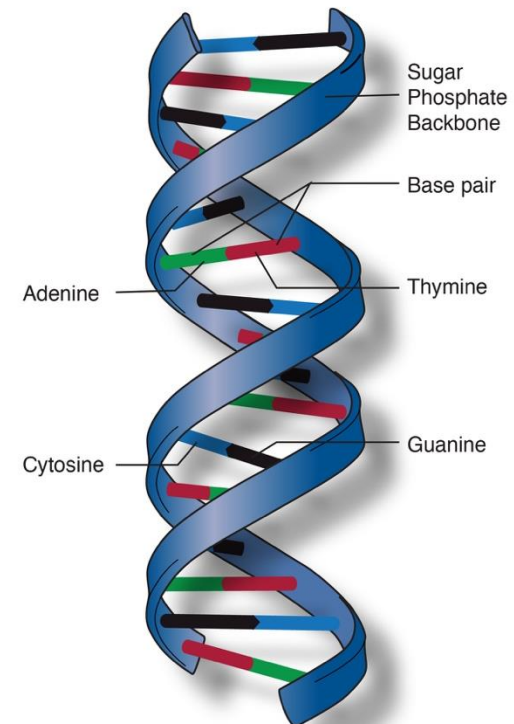
A binds to T, C binds to G

Forward strand: **GATTACAT**

Reverse strand: **CTAATGTA**

Is a palindrome: **ACGTACGT**
TGCATGCA

Is not a palindrome: **ACGTTGCA**
TGCAACGT



Assignment 4 released

Assignment 4: RNA-seq and Machine Learning

Assignment Date: Wednesday, October 15, 2025

Due Date: Wednesday, October 22 @ 11:59pm

Assignment Overview

In this assignment you will explore a couple of key aspects of RNA-seq and introduce the key concepts of machine learning. For this assignment, we will provide a Jupyter notebook with code for you to use and complete your assignment in.

To use the Jupyter notebook, create a new mamba environment for assignment 4:

```
mamba create -n asn4 scikit-learn pytorch matplotlib jupyter umap-learn seaborn pandas numpy
```

Make sure to download the training data for the assignment from [Dropbox](#). Place this in the `data_files` directory and make sure to uncompress (`gunzip`) both the training and testing files.

As a reminder, any questions about the assignment should be posted to [Piazza](#).

See the notebook here: [Assignment4.ipynb](#)

Packaging

The solutions to the above questions should be submitted as a single PDF document that includes your name, email address, and all relevant figures (as needed). If you use ChatGPT for any of the code, also record the prompts used. Submit your solutions by uploading the PDF to [GradeScope](#), and remember to select where in your submission each question/subquestion is. The Entry Code is: GVXGV2.

If you submit after this time, you will use your late days. Remember, you are only allowed 4 late days for the entire semester!

Resources

- Jupyter notebooks: <https://jupyter.org/>
- scikit-learn: <https://scikit-learn.org/stable/>
- pytorch: <https://pytorch.org/>
- umap-learn: https://umap-learn.readthedocs.io/en/latest/basic_usage.html

Lecture 15. Single Cell Genomics

Michael Schatz

Mahler Revsine

October 15, 2025

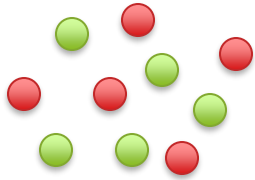
Applied Comparative Genomics



Population Heterogeneity

Red cells express twice the abundance of “brain” genes compared to green cells

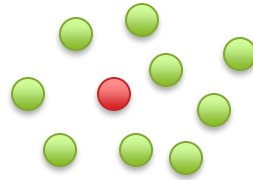
Experiment 1: 50/50



Compared to a control sample of pure green cells, this sample will show:

$50\% \ 2x + 50\% \ 1x$
 $= 1.5x$ over expression
of brain genes

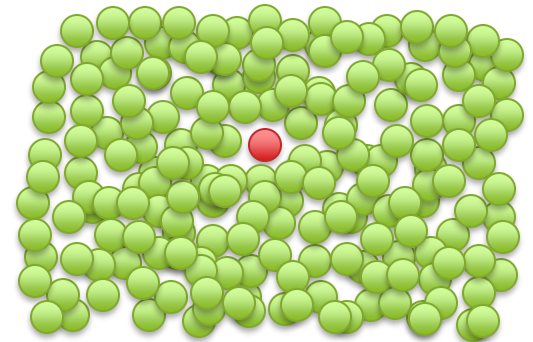
Experiment 2: 1/10



Compared to a control sample of pure green cells, this sample will show:

$10\% \ 2x + 90\% \ 1x$
 $= 1.1x$ over expression
of brain genes

Experiment 3: 1/1000



Compared to a control sample of pure green cells, this sample will show:

$0.1\% \ 2x + 99.1\% \ 1x$
 $= 1.001x$ over expression
of brain genes

The limitations of averages

	Drug A	Drug B
Overall Response	78% (273/350)	83% (289/350)

The limitations of averages

	Drug A	Drug B
Overall Response	78% (273/350)	83% (289/350)
Male Response	93% (81/87)	87% (234/270)
Female Response	73% (192/263)	69% (55/80)

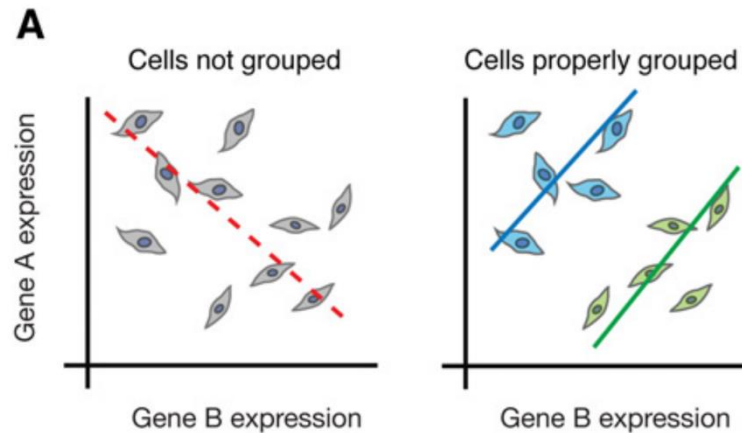
What??? How can the better performing drug depend on if you examine the overall response or separately examine by gender

Example of Simpson's paradox:

Trend of the overall average may reverse the trends of each constituent group

In this example, the "lurking" variable (or confounding variable) is the severity of the case (represented by the doctors' treatment decision trend of favoring B for less severe cases), which was not previously known to be important until its effects were included. (Based on real analysis of kidney stone treatments)

The paradox of averages



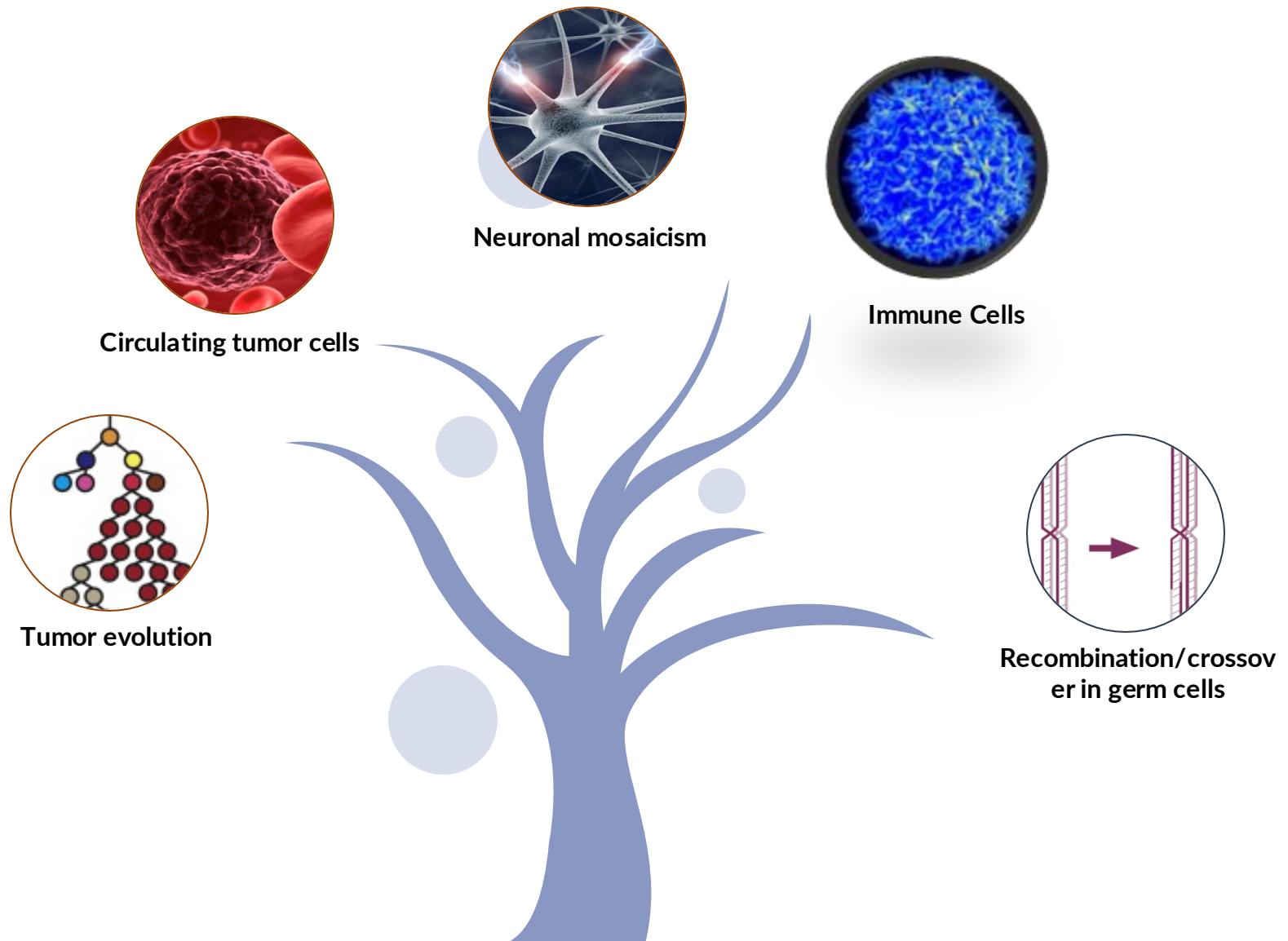
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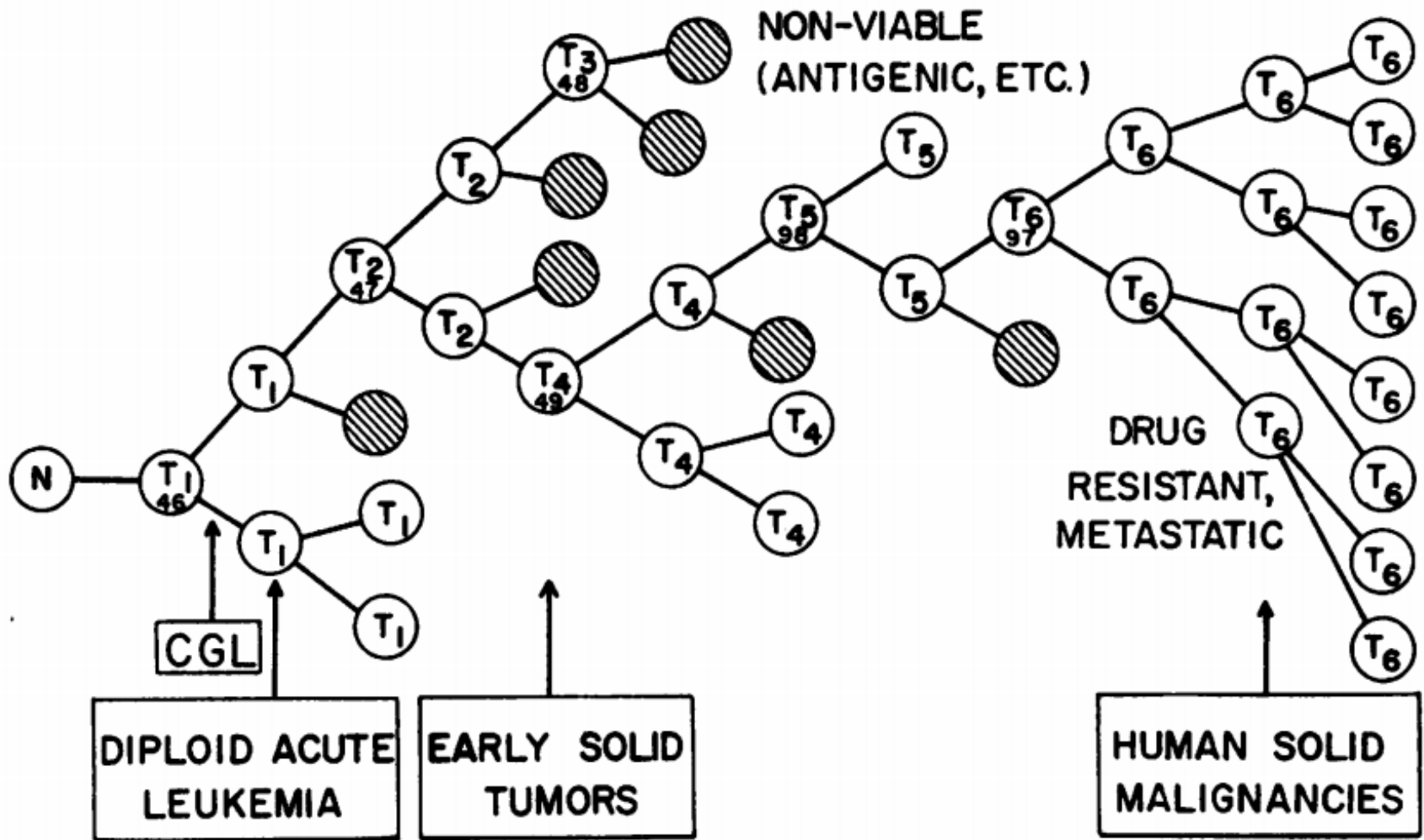
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Sources of (Genomic) Heterogeneity

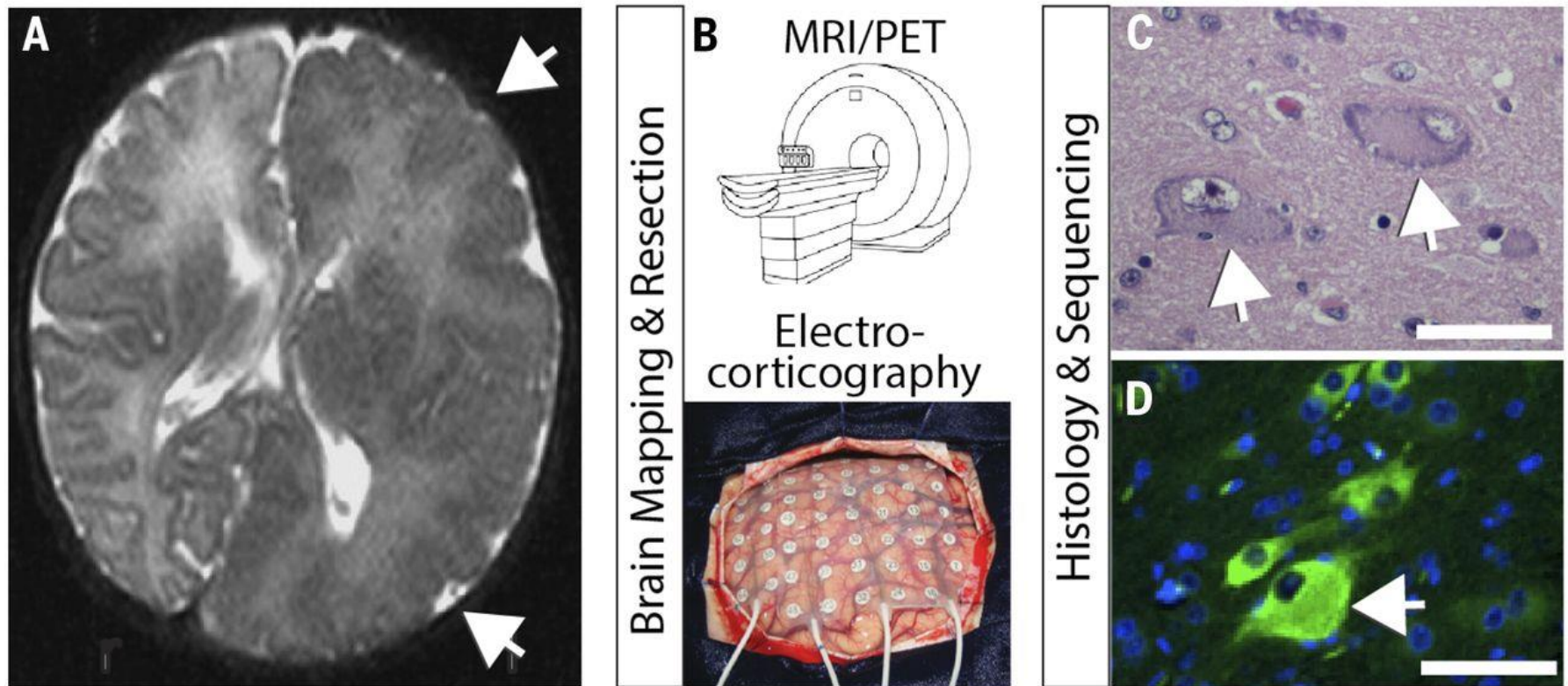


Tumor Evolution



The Clonal Evolution of Tumor Cell Populations

Peter C. Nowell (1976) *Science*. 194(4260):23-28 DOI: 10.1126/science.959840

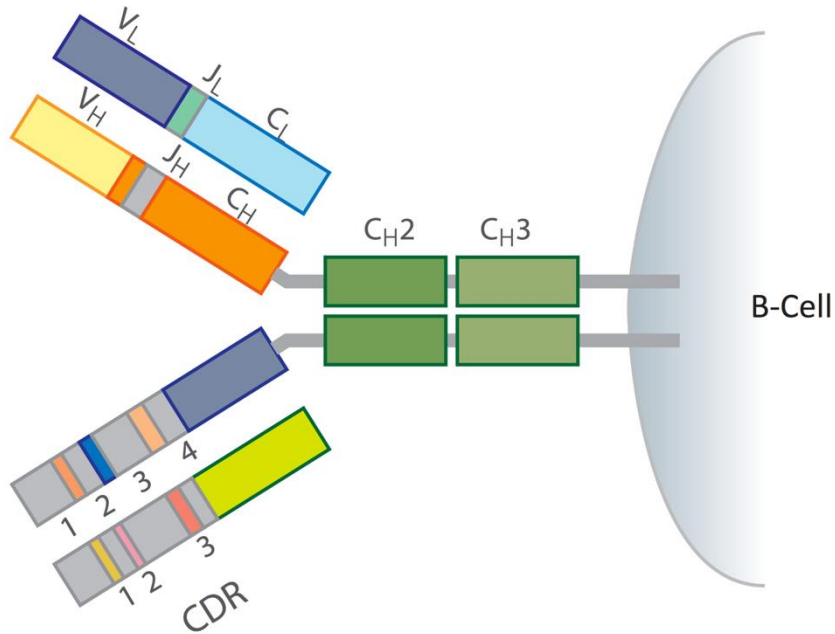


An example of brain somatic mosaicism that leads to a focal overgrowth condition.

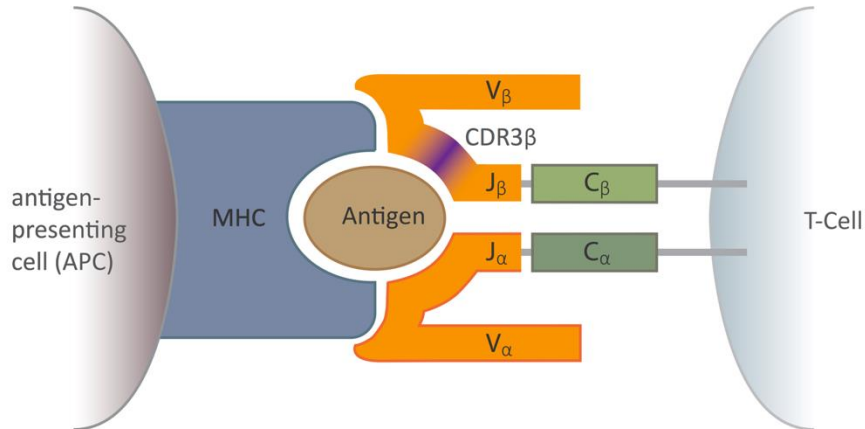
(A) Axial brain MRI of focal overgrowth from a 2-month-old child with intractable epilepsy and intellectual disability. **(B)** Brain mapping using high-resolution MRI is followed by surgical resection of diseased brain tissue. **(C)** Histological analysis with hematoxylin/eosin showing characteristic balloon cells consisting of large nuclei, distinct nucleoli, and glassy eosinophilic cytoplasm. **(D)** After surgery, the patient showed clinical improvement.

Intersection of diverse neuronal genomes and neuropsychiatric disease: The Brain Somatic Mosaicism Network. McConnell et al (2017) Science. doi: 10.1126/science.aal1641

Immunology

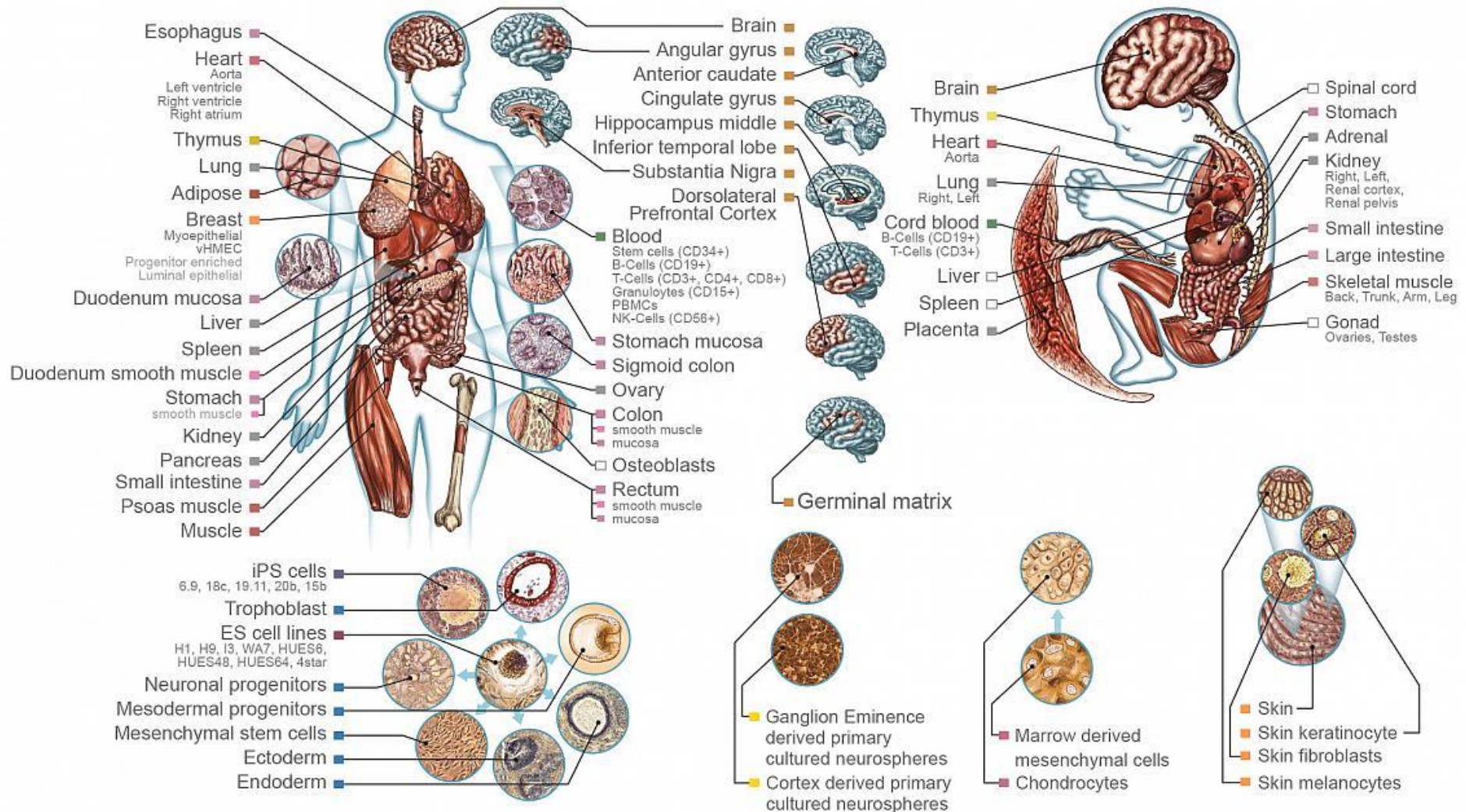


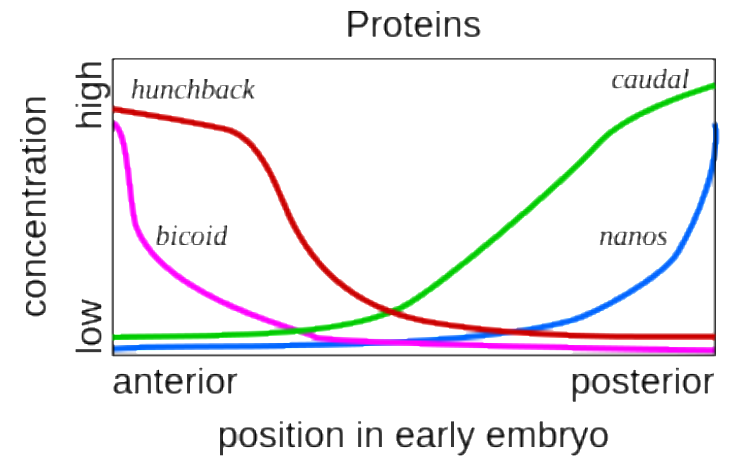
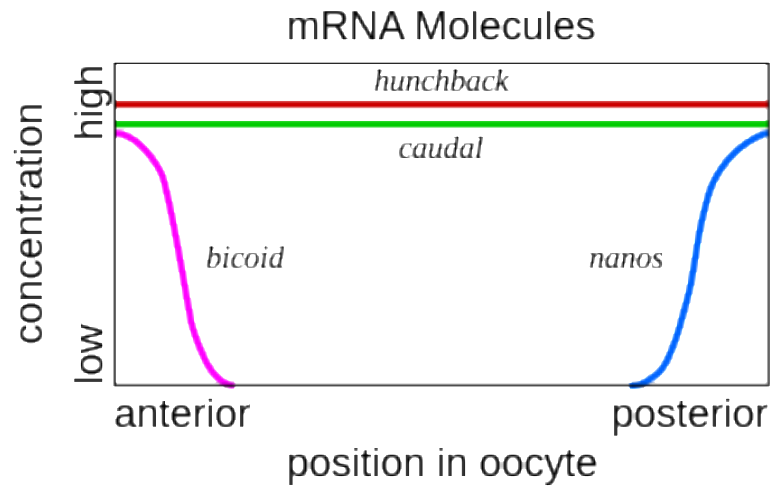
- Massive diversity rivaled only by germ cells
- Somatic recombination



- B cells – antibody generation
- T cells – antigen response

Sources of (Cellular) Heterogeneity



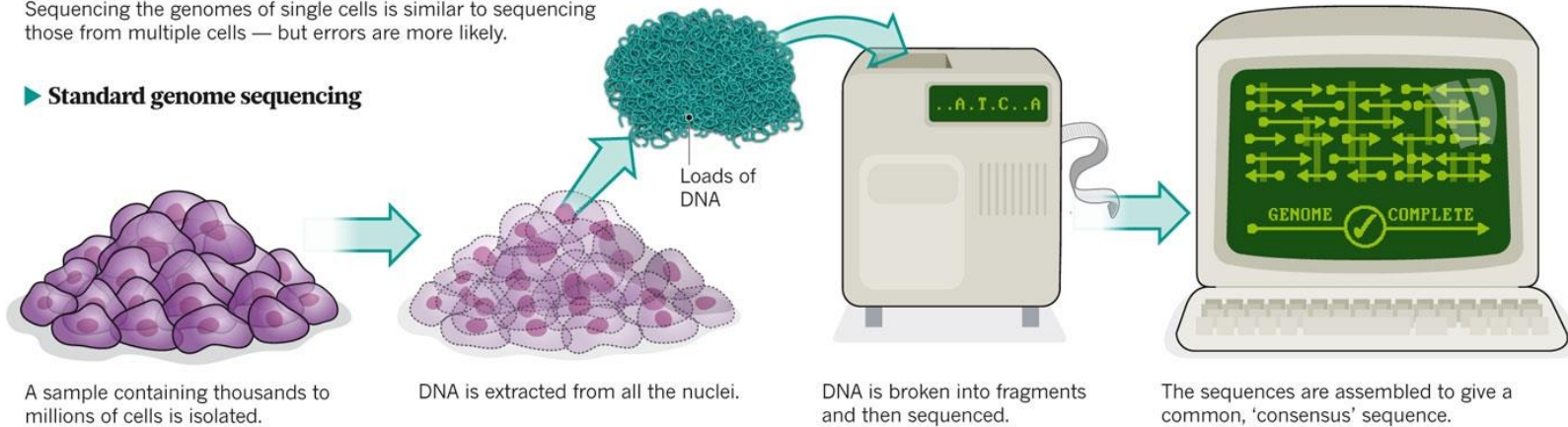


Single-cell vs. bulk sequencing

ONE GENOME FROM MANY

Sequencing the genomes of single cells is similar to sequencing those from multiple cells — but errors are more likely.

► Standard genome sequencing

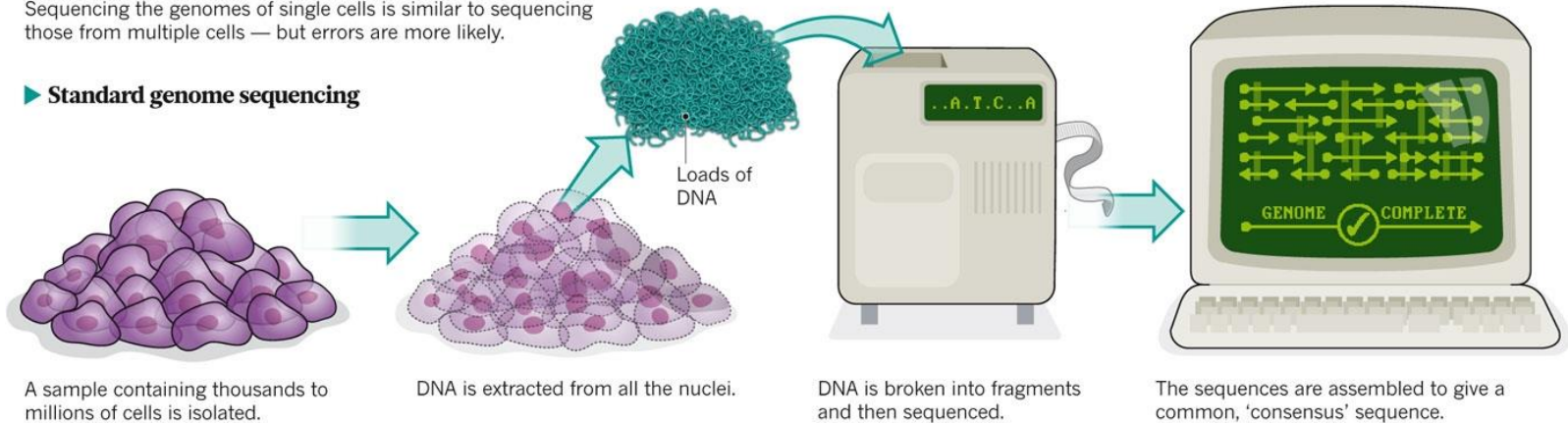


Single-cell vs. bulk sequencing

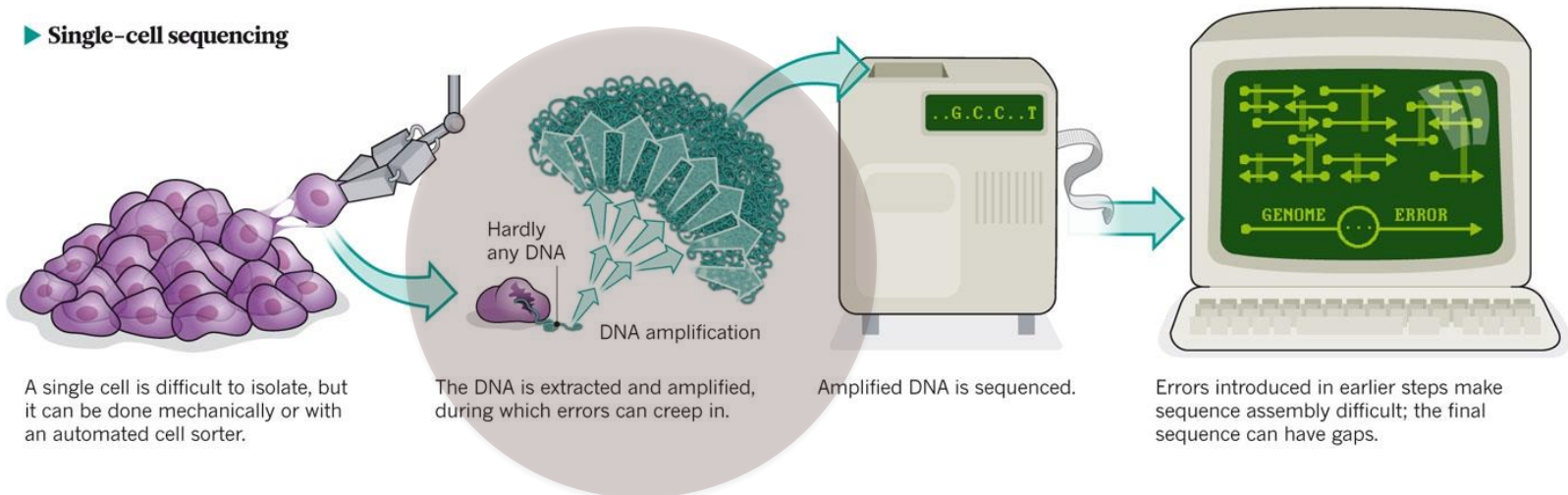
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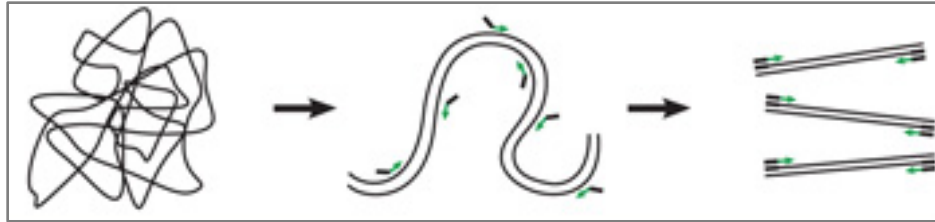
► Standard genome sequencing



► Single-cell sequencing

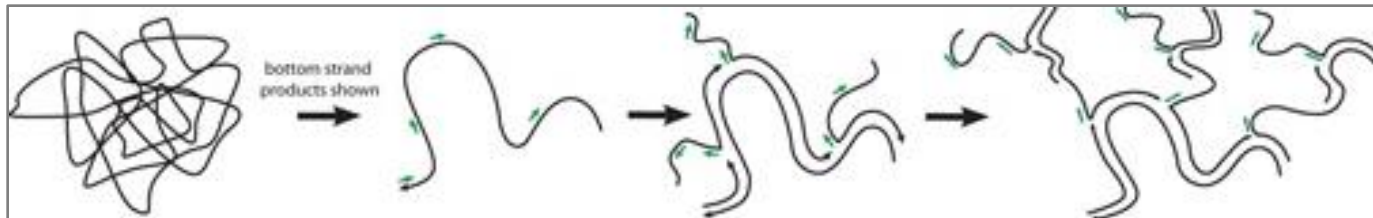


Whole Genome Amplification Techniques



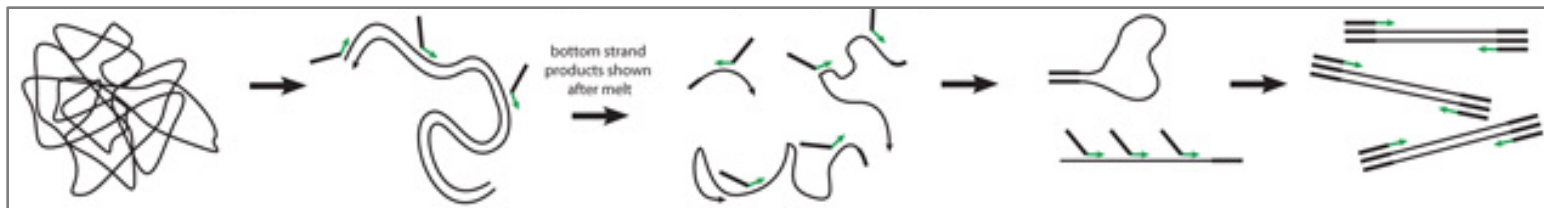
DOP-PCR: Degenerate Oligonucleotide Primed PCR

Telenius et al. (1992) Genomics



MDA: Multiple Displacement Amplification

Dean et al. (2002) PNAS

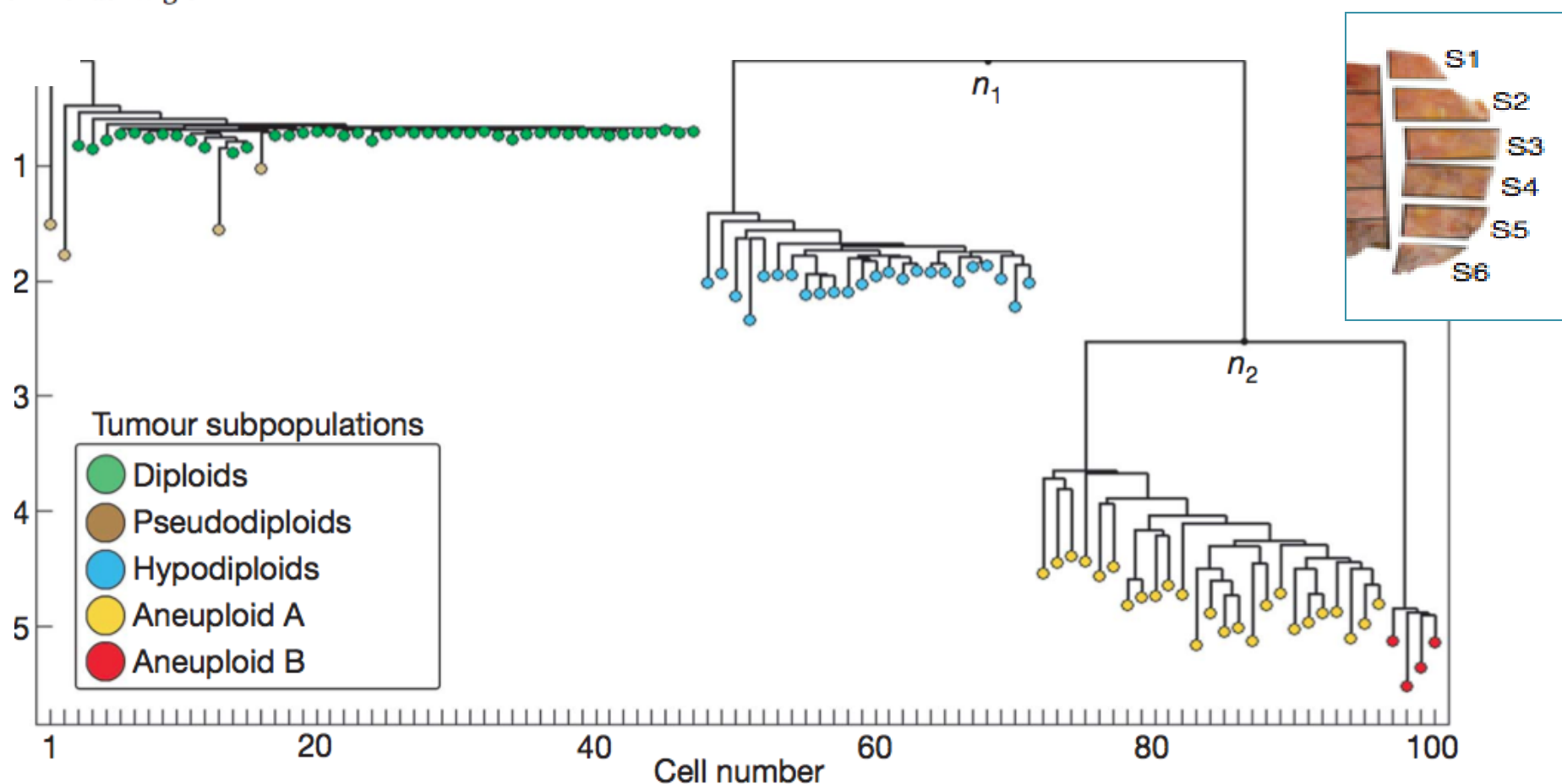


MALBAC: Multiple Annealing and Looping Based Amplification Cycles

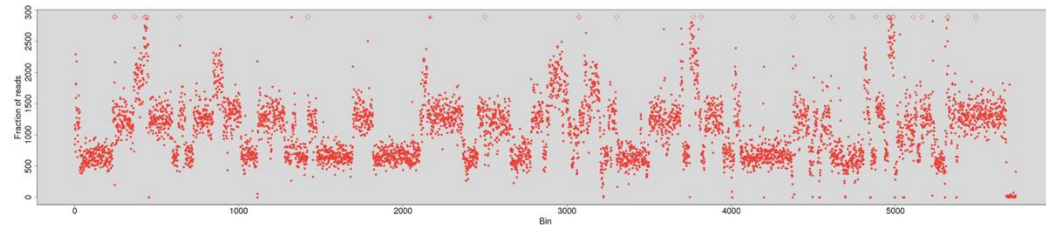
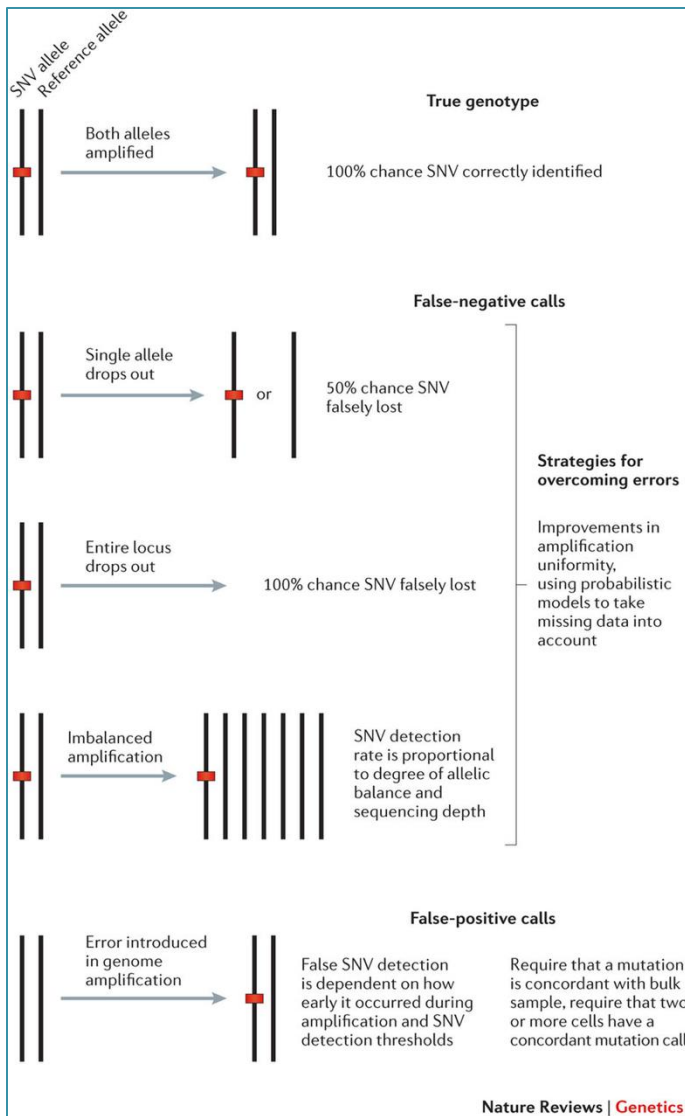
Zong et al. (2012) Science

Tumour evolution inferred by single-cell sequencing

Nicholas Navin^{1,2}, Jude Kendall¹, Jennifer Troge¹, Peter Andrews¹, Linda Rodgers¹, Jeanne McIndoo¹, Kerry Cook¹, Asya Stepansky¹, Dan Levy¹, Diane Esposito¹, Lakshmi Muthuswamy³, Alex Krasnitz¹, W. Richard McCombie¹, James Hicks¹ & Michael Wigler¹



scCNVs



Potential for biases at every step

- WGA: Non-uniform amplification
- Library Preparation: Low complexity, read duplications, barcoding
- Sequencing: GC artifacts, short reads
- Computation: mappability, GC correction, segmentation, tree building

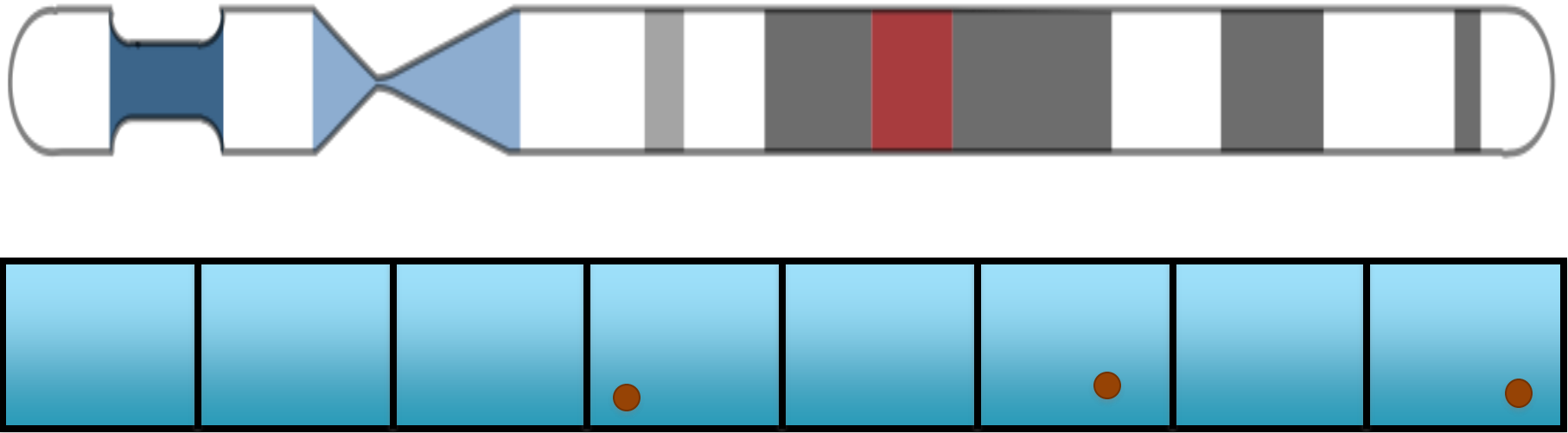
Coverage is very sparse and noisy

-> requires special processing

Single-cell genome sequencing: current state of the science

Gawad et al (2016) Nature Reviews Genetics. doi:10.1038/nrg.2015.16

I) Binning

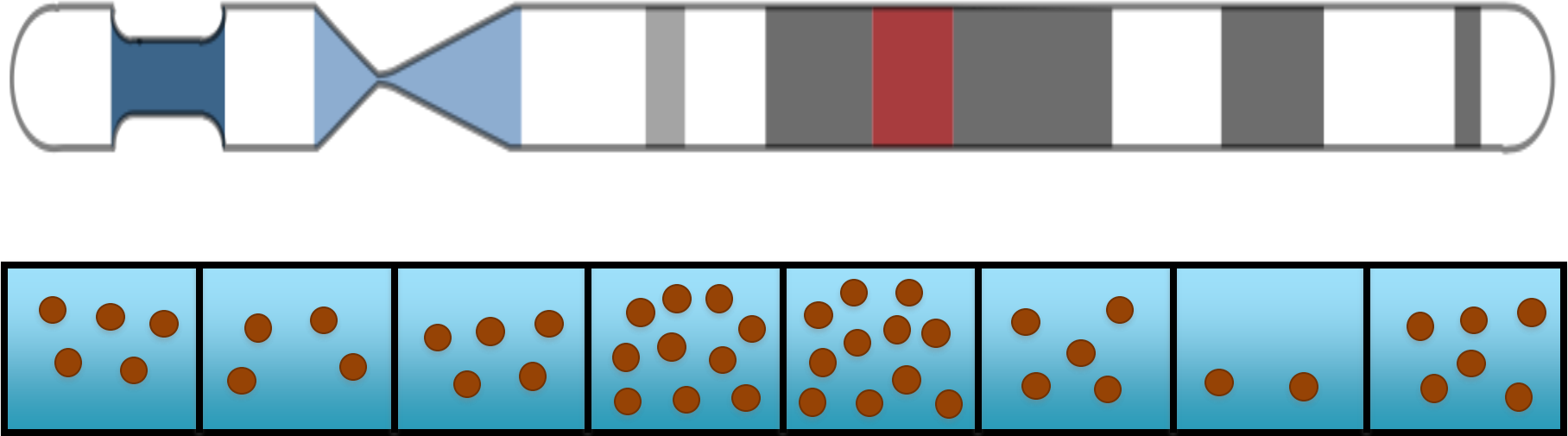


Single Cell CNV analysis

- Divide the genome into “bins” with $\sim 50 - 100$ reads / bin
- Map the reads and count reads per bin

Use uniquely mappable bases to establish bins

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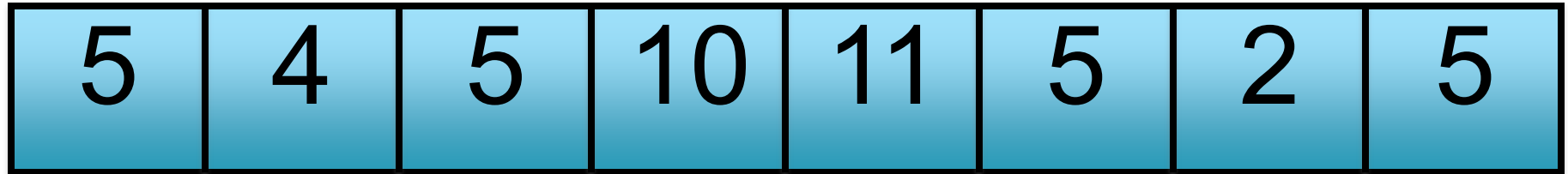


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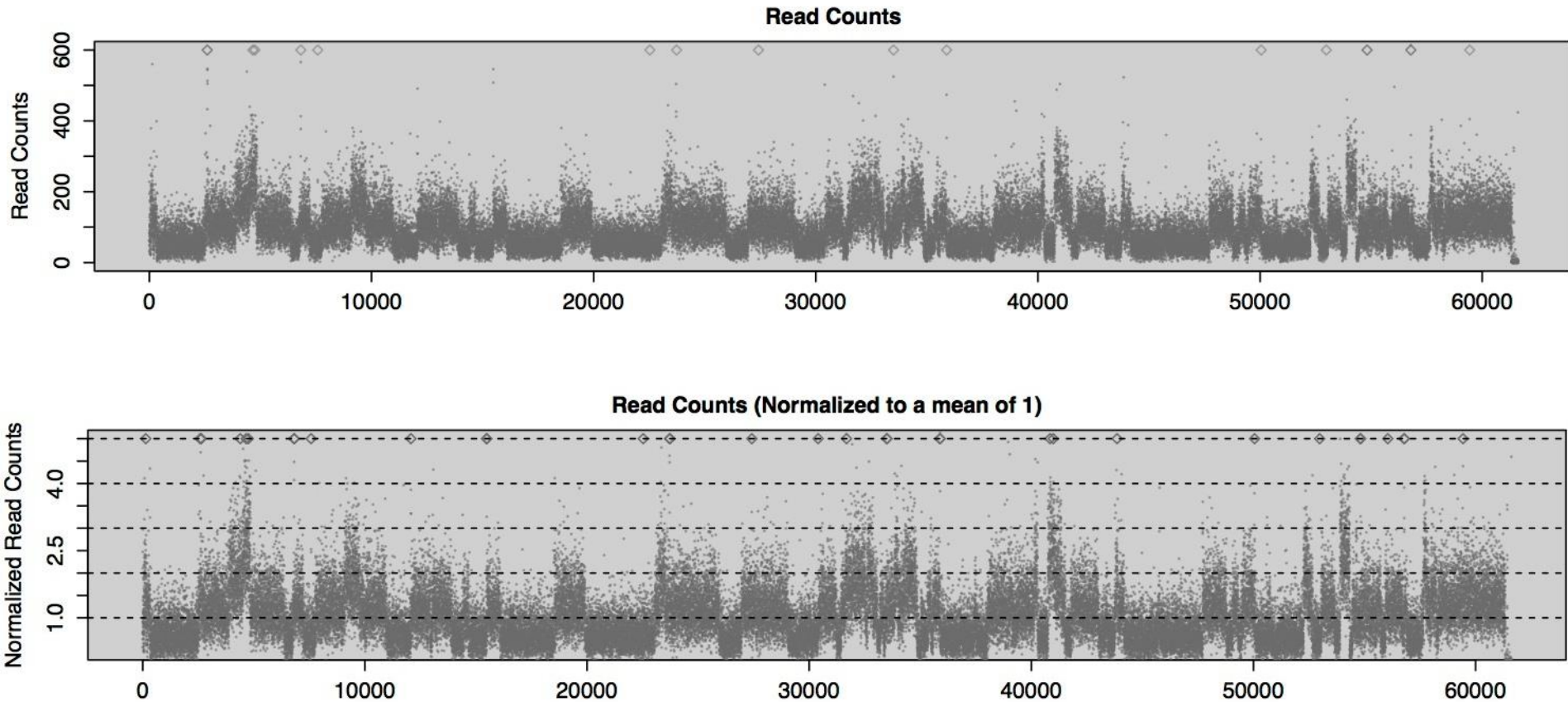


Single Cell CNV analysis

- Divide the genome into “bins” with ~50 – 100 reads / bin
- Map the reads and count reads per bin

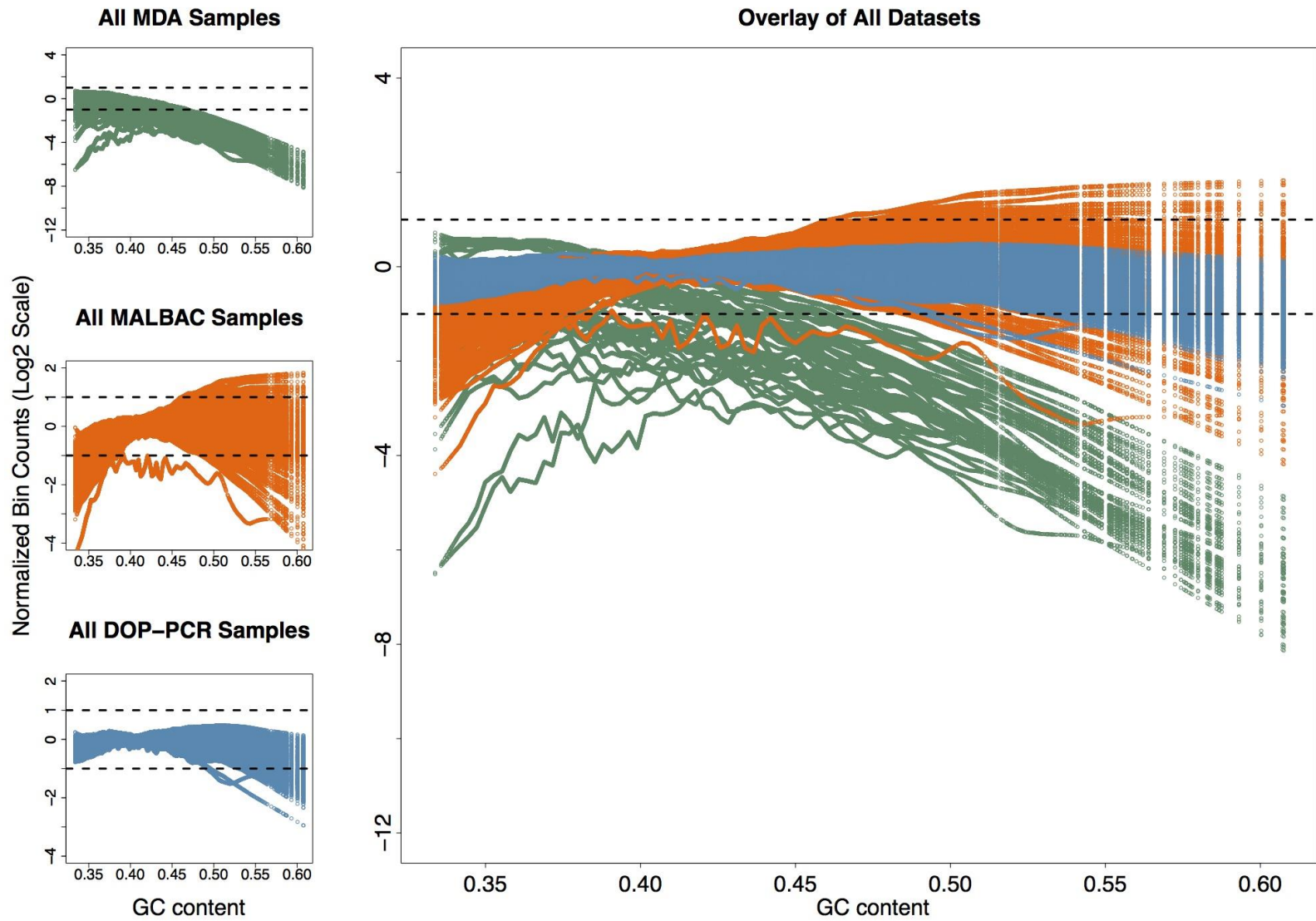
Use uniquely mappable bases to establish bins

2) Normalization

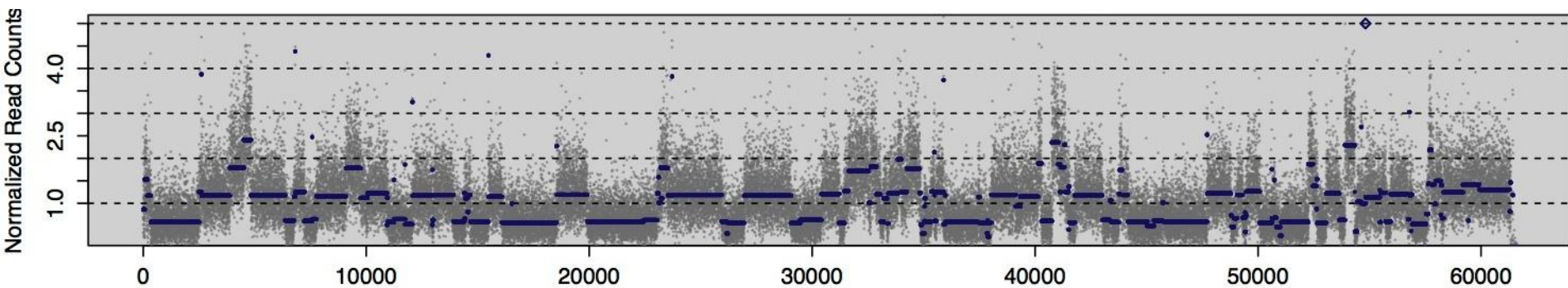


Also correct for mappability, GC content, amplification biases

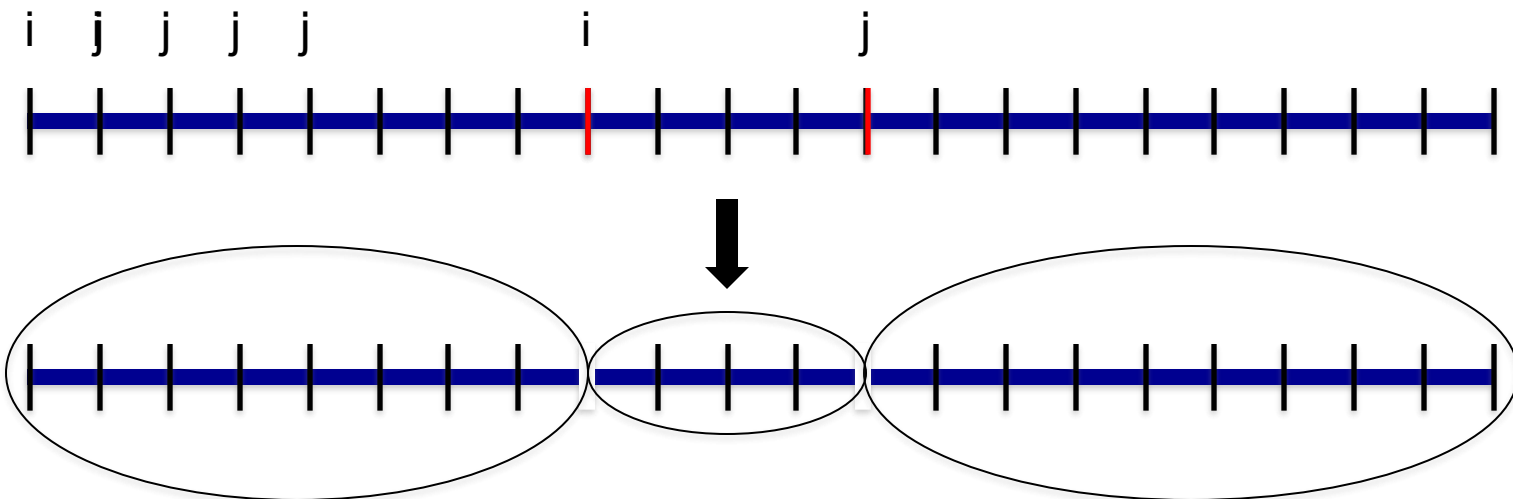
GC Bias



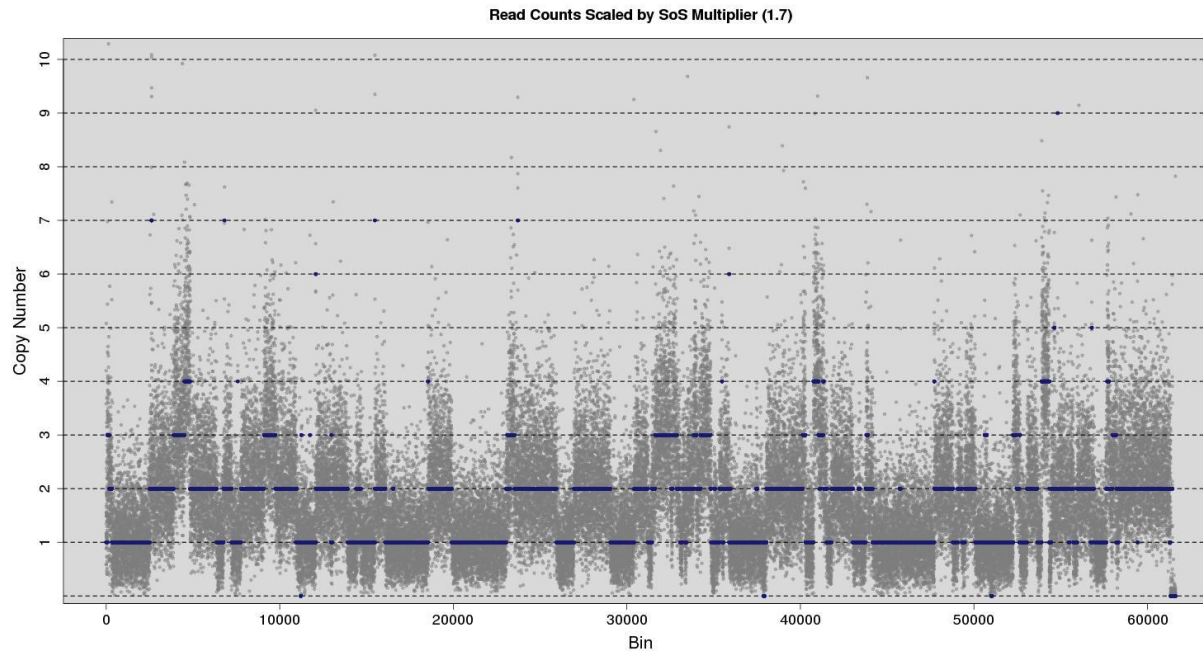
3) Segmentation



Circular Binary Segmentation (CBS)

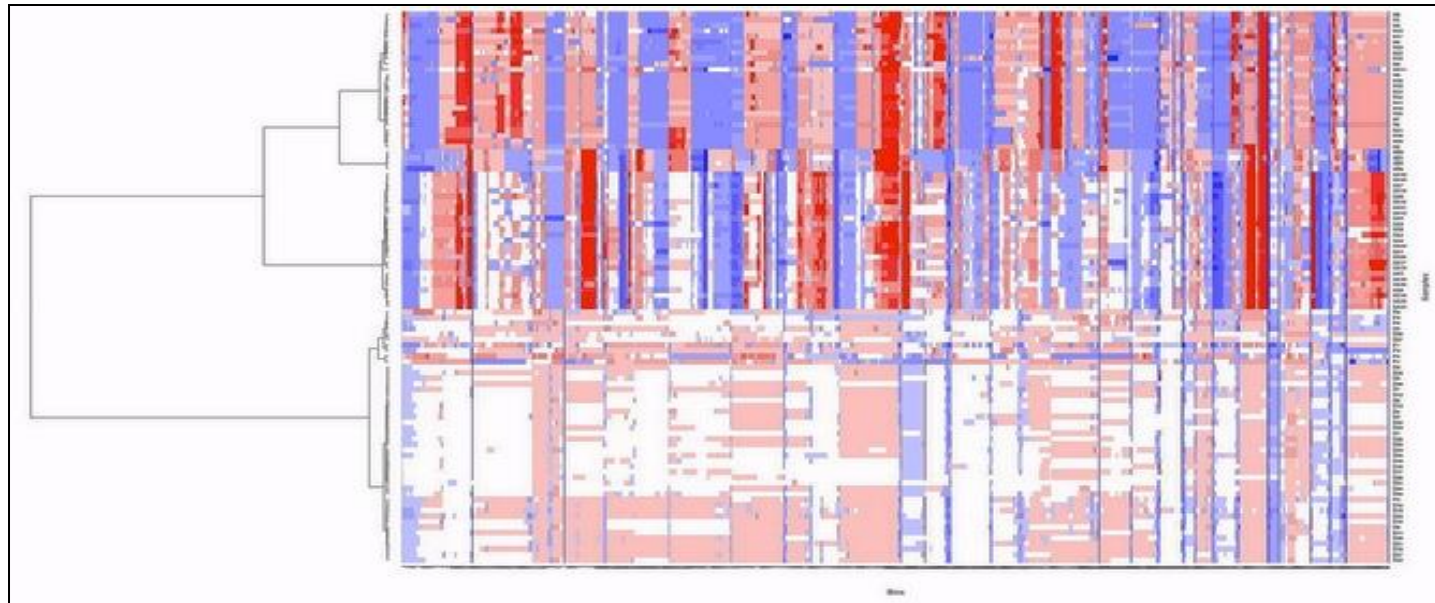
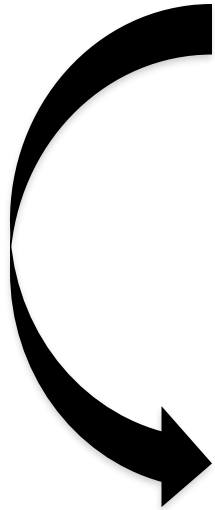
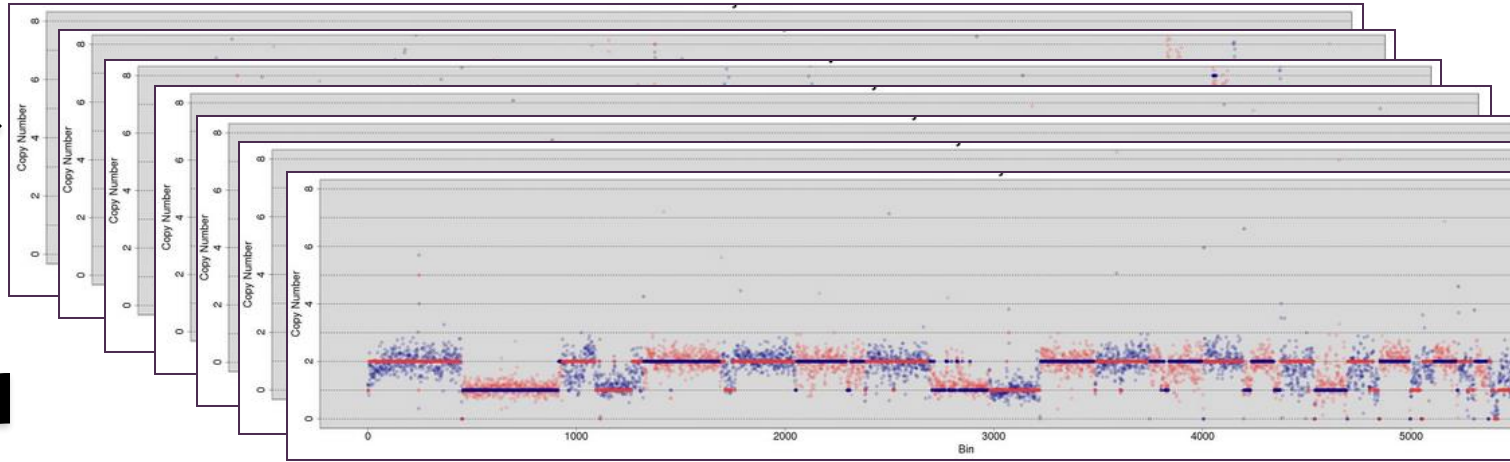
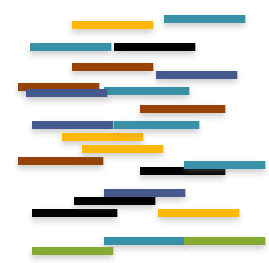


4) Estimating Copy Number



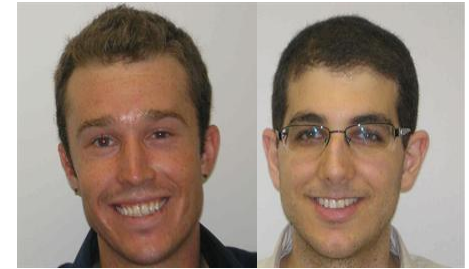
$$CN = \operatorname{argmin} \left\{ \sum_{i,j} (\hat{Y}_{i,j} - Y_{i,j})^2 \right\}$$

5) Cells to Populations



Ginkgo

<http://qb.cshl.edu/ginkgo>



Interactive Single Cell CNV analysis & clustering

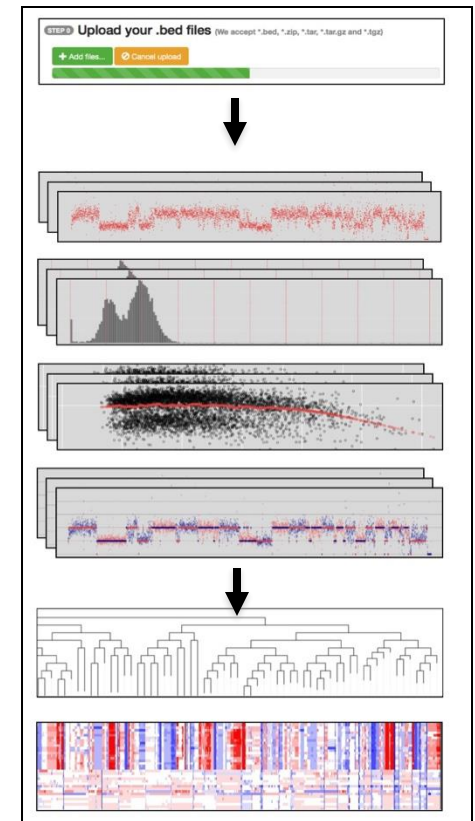
- Easy-to-use, web interface, parameterized for binning, segmentation, clustering, etc
- Per cell through project-wide analysis in any species

Compare MDA, DOP-PCR, and MALBAC

- DOP-PCR shows superior resolution and consistency

Available for collaboration

- Analyzing CNVs with respect to different clinical outcomes
- Extending clustering methods, prototyping scRNA



Interactive analysis and assessment of single-cell copy-number variations.

Garvin T, Aboukhalil R, Kendall J, Baslan T, Atwal GS, Hicks J, Wigler M, Schatz MC (2015)

Nature Methods doi:10.1038/nmeth.3578



Single Cell CNV-Seq

- Reveal genomic heterogeneity
- Understand clonal evolution
- Determine pathogenesis and cancer progression
- Scalable from 100s-1000s of cells
- Single-cell CNV calling
- Call CNVs down to 100kb resolution
- CNV-Seq specific software pipeline

Single-cell RNA sequencing, “the bioinformatician’s microscope”

— a snapshot of the underlying biology in a data matrix.

Brain cells



Biological sample

number of times **gene i**
was expressed in **cell j**



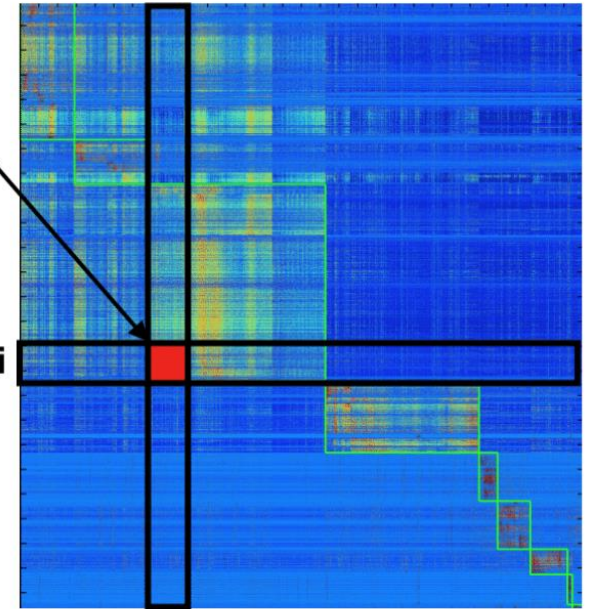
sequencing

Genes (features)

Single Cells (samples)

cell j

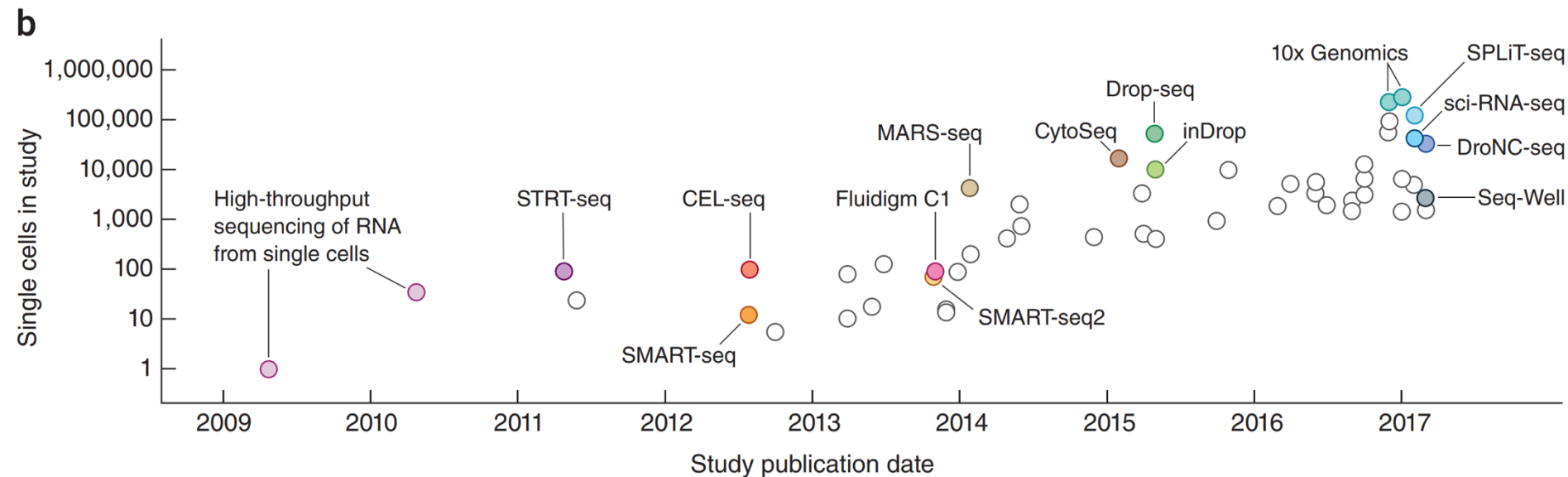
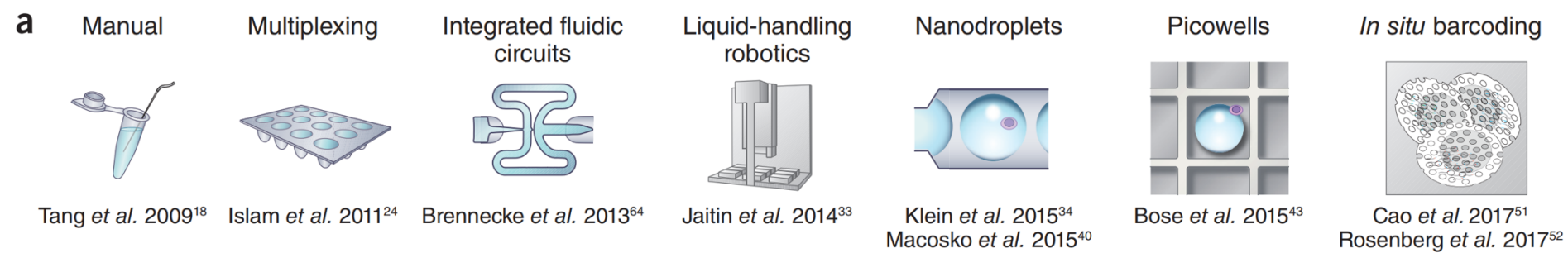
gene i

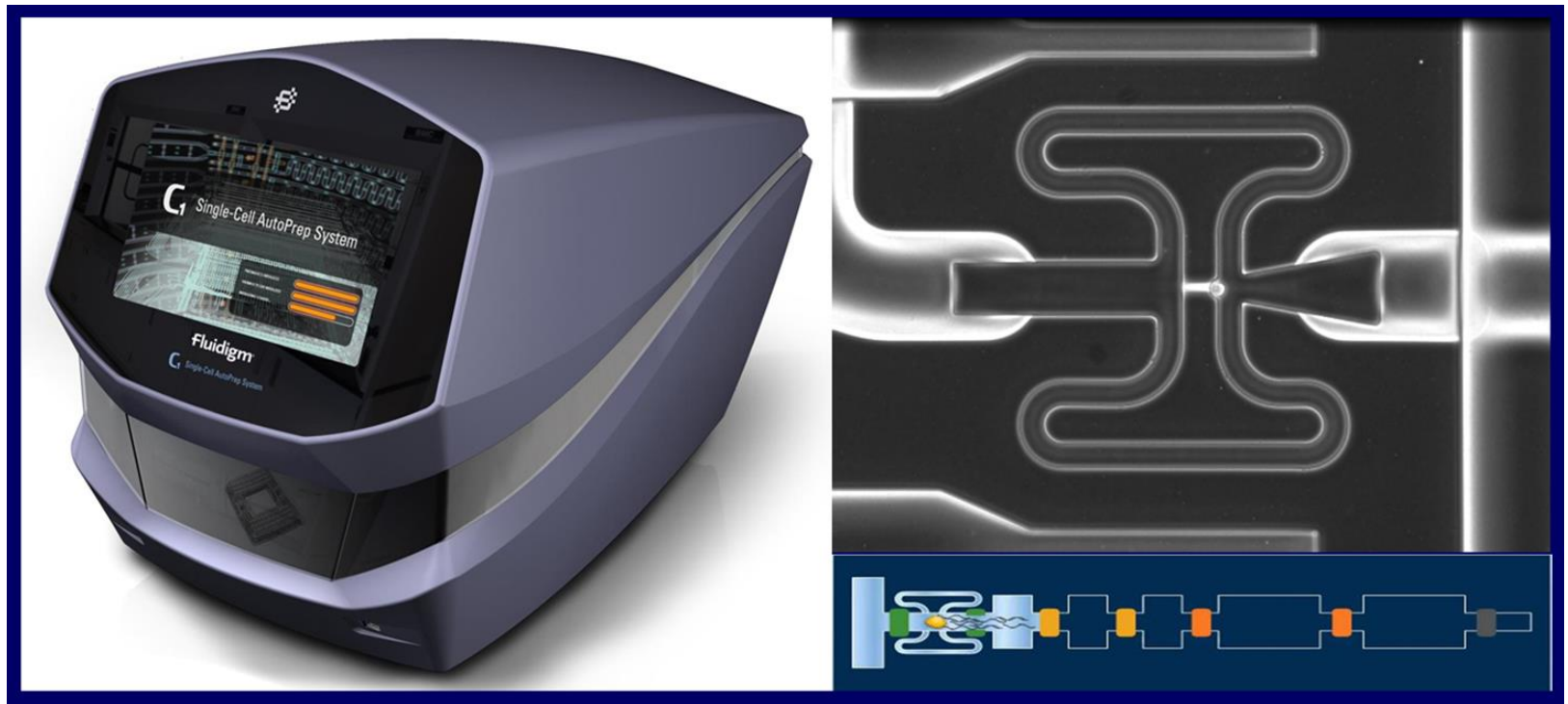


Gene expression matrix

computationally explore complex biological systems

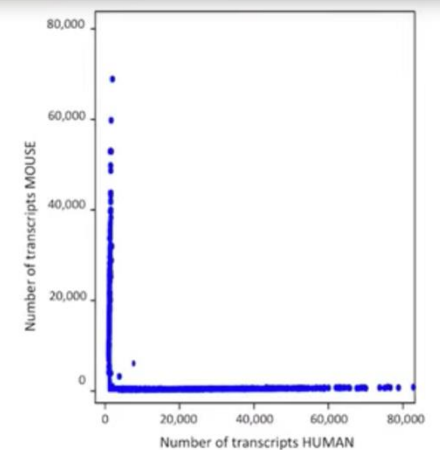
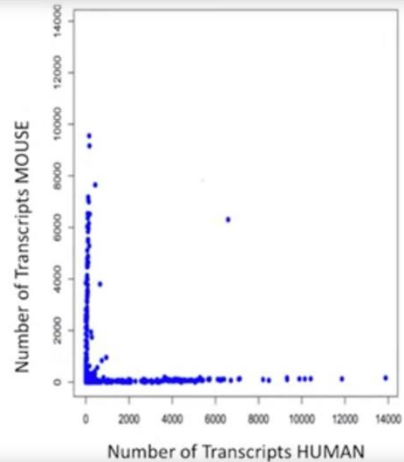
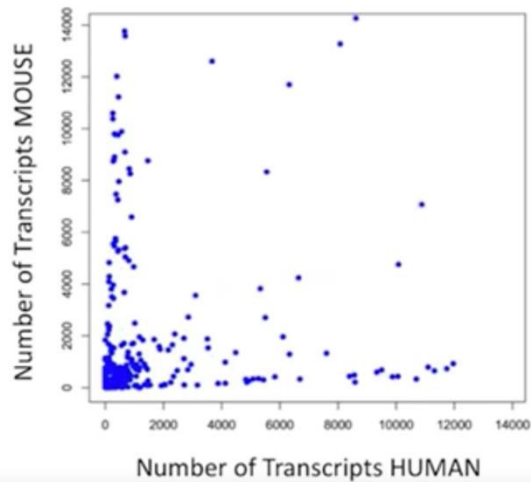
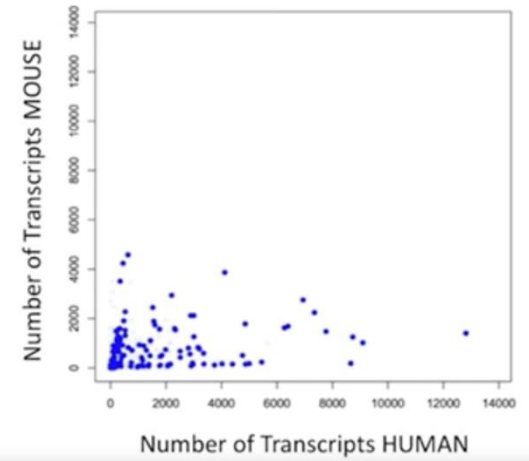
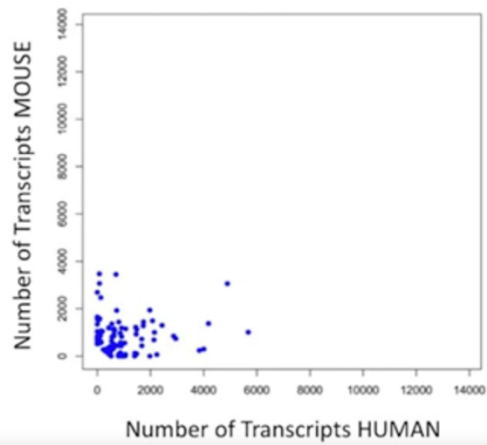
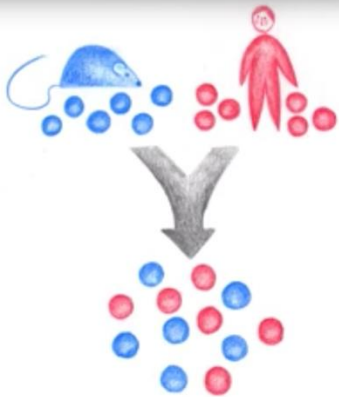
A decade of single-cell RNA-seq





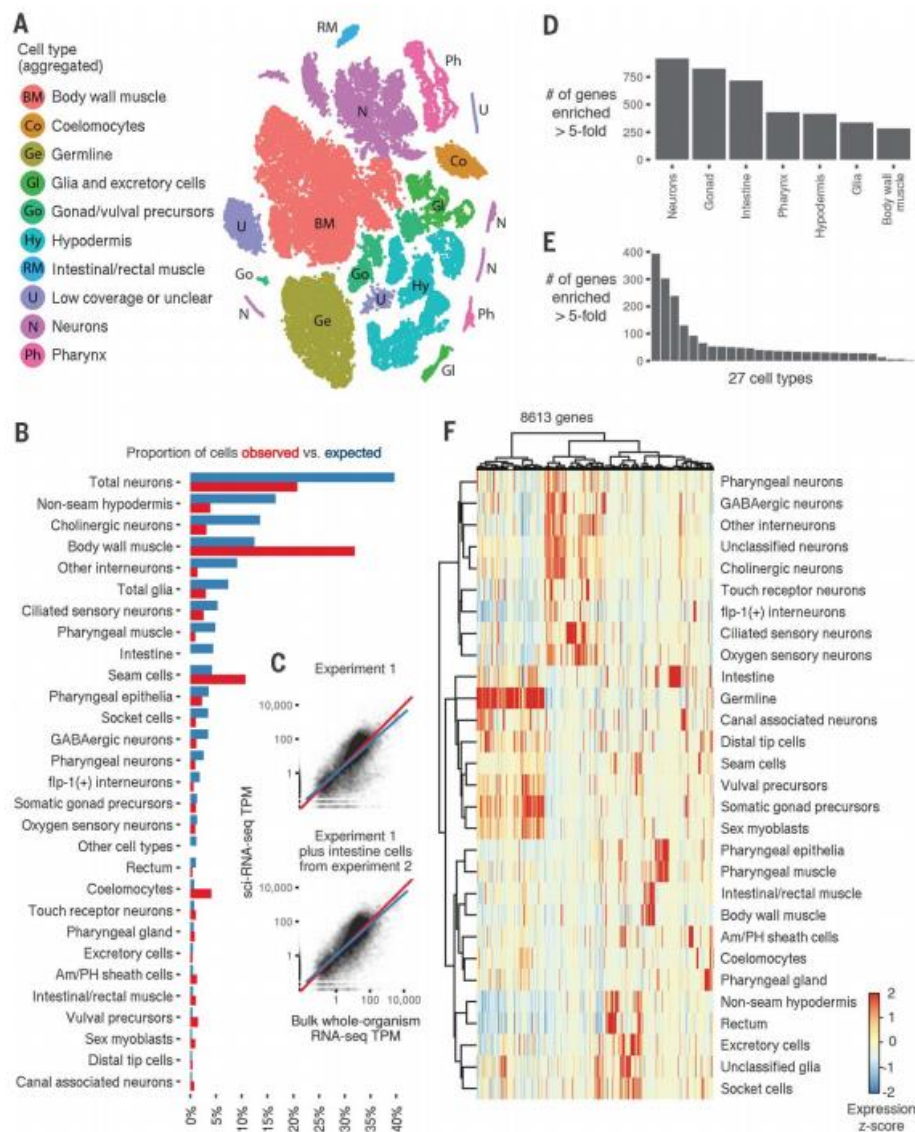
Fluidigm C1

Benchtop automated single-cell isolation and preparation system(lysis and pre-amplification) for genomic analysis. The C1 System provides an easy and highly reproducible workflow to process **96 single cells** for DNA or RNA analysis.



Drop-seq: Droplet barcoding of single cells

<https://www.youtube.com/watch?v=vL7ptq2Dcf0>



Key Results

Profile every cell of *C. elegans* larva using combinatorial indexing

(a) t-SNE visualization of clusters

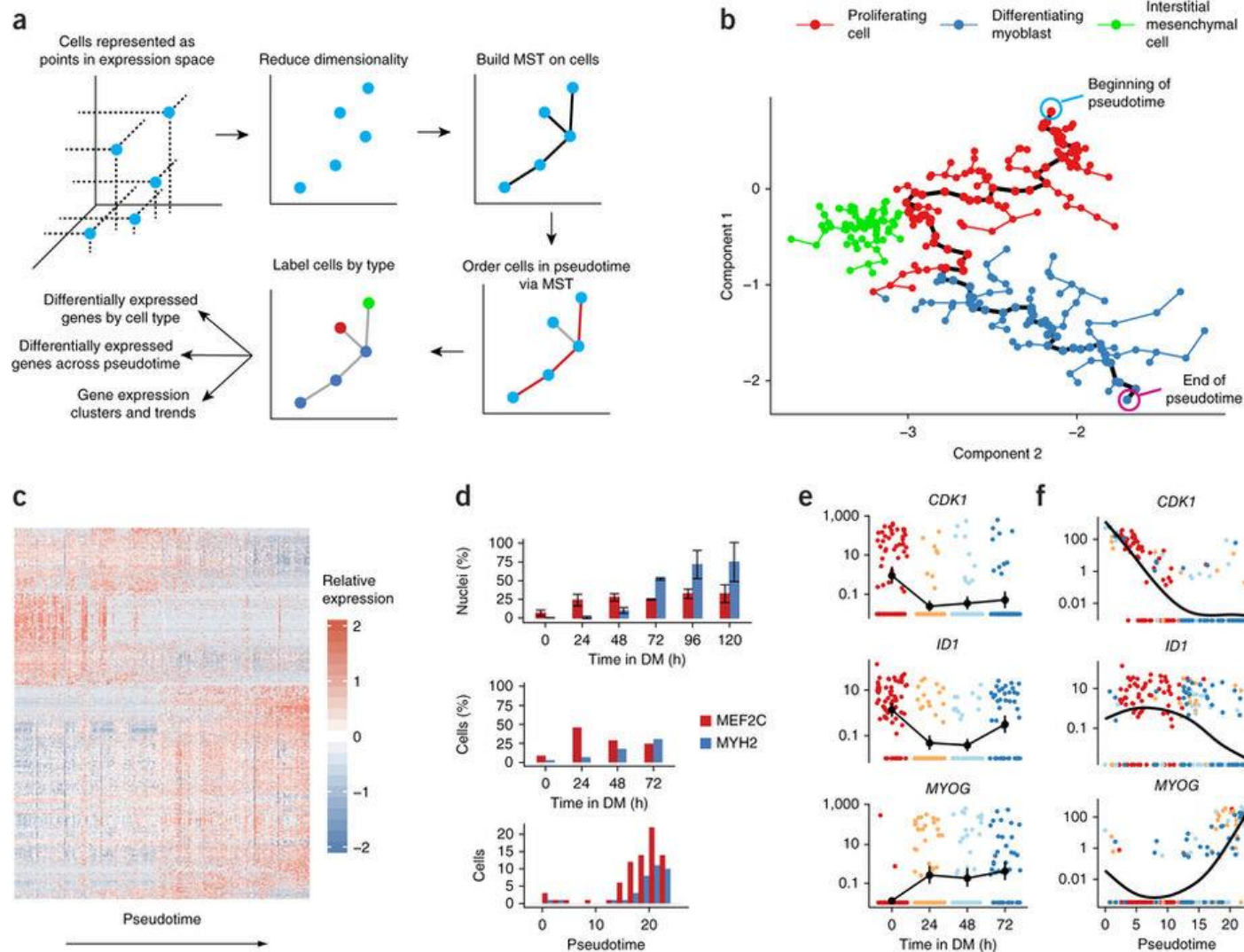
(b) Proportion of cells observed vs expected match well (including cells that only occur once or twice in the animal)

(c) Good correlation between single cell and bulk analysis of selected cell types

(d-f) Analysis of key genes per cell type

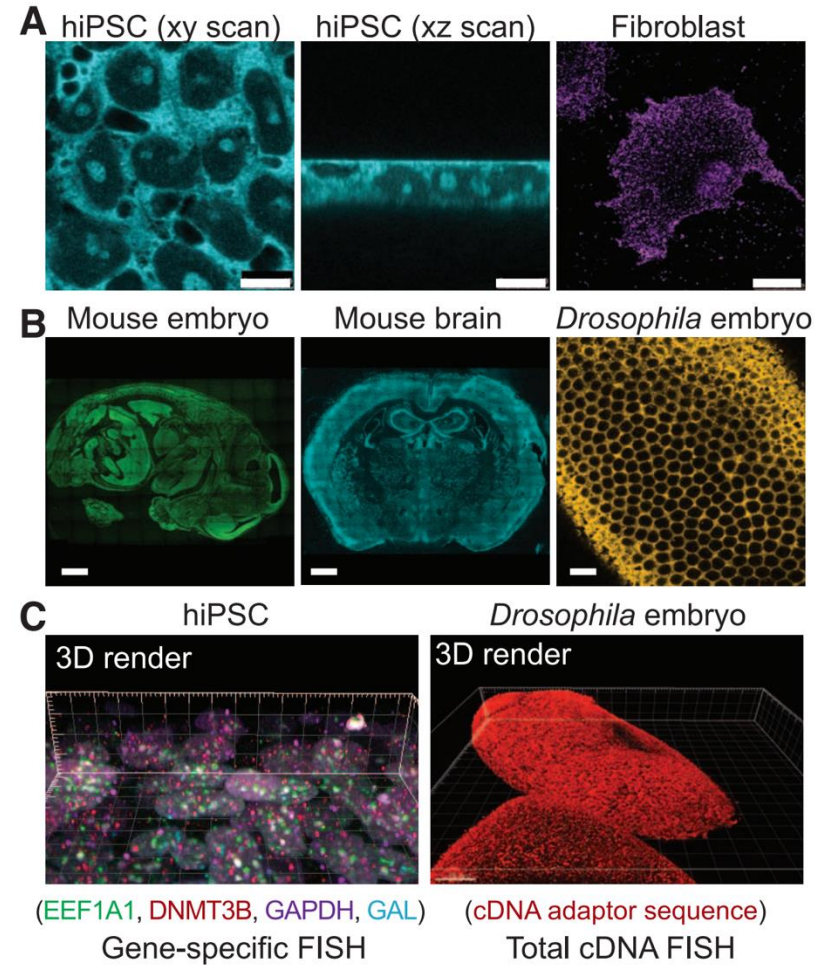
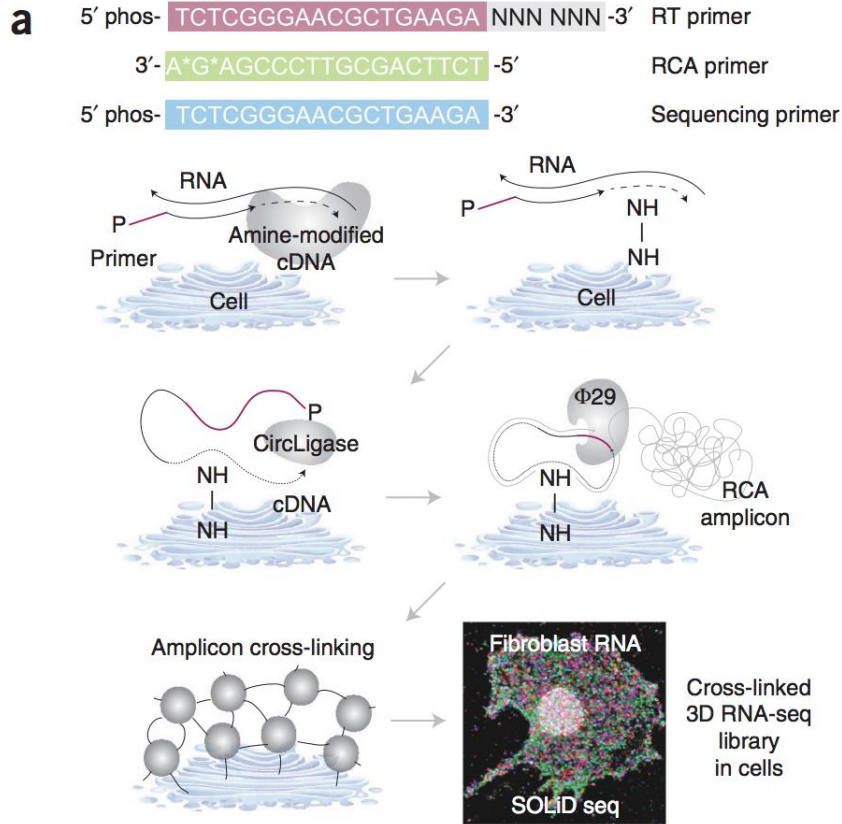
Comprehensive single-cell transcriptional profiling of a multicellular organism

Cao et al (2017) Science. 357:661-557



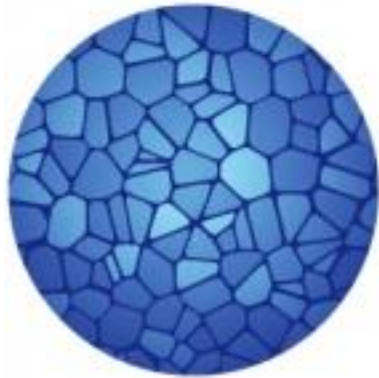
The dynamics and regulators of cell fate decisions are revealed by pseudotemporal ordering of single cells (“Monocle”)

Trapnell et al (2014) Nature Biotechnology. doi:10.1038/nbt.2859

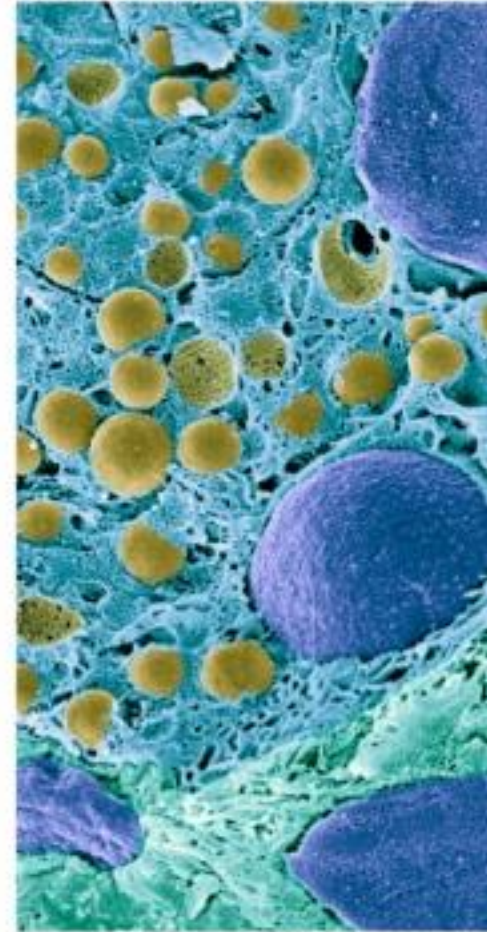
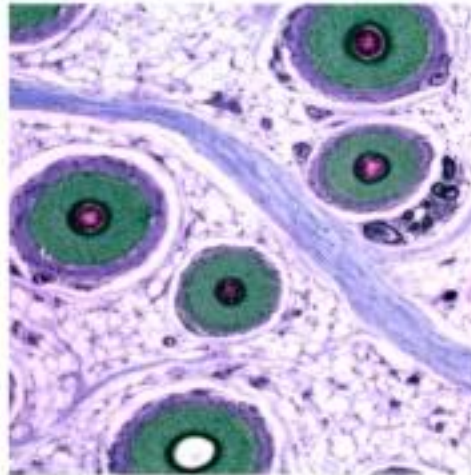


Highly multiplexed subcellular RNA sequencing in situ (“FISSEQ”)

Lee et al (2014) Science. doi: 10.1126/science.1250212



HUMAN CELL ATLAS



<https://www.humancellatlas.org/>

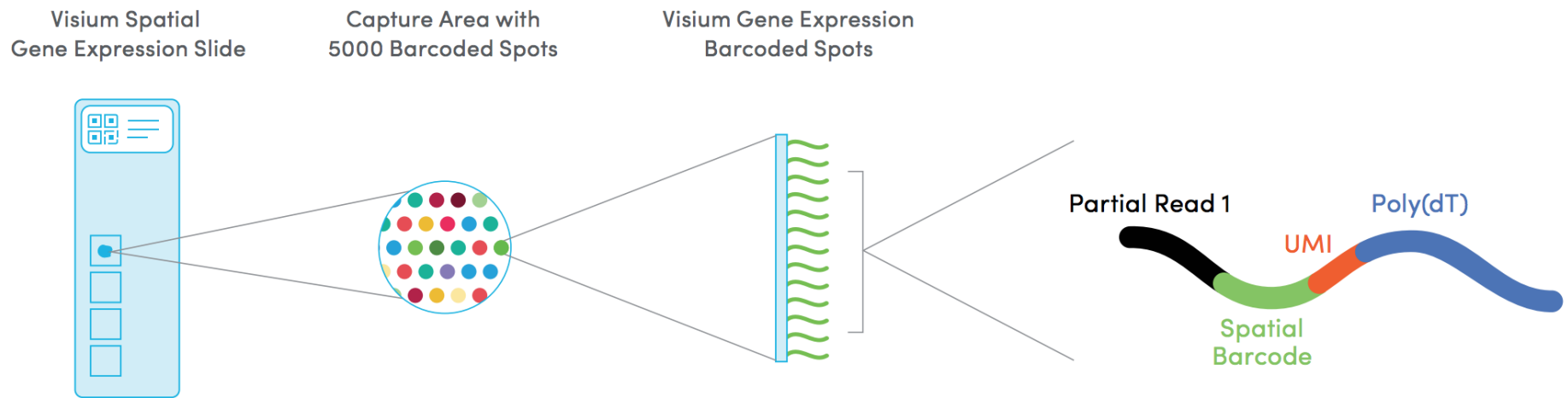


Figure 1. Here we show the composition of the Visium Spatial Gene Expression slide. Each slide contains four Capture Areas with approximately 5000 barcoded spots, which in turn contain millions of spatially-barcoded capture oligonucleotides. Tissue mRNA is released and binds to the barcoded oligos, enabling capture of gene expression information.

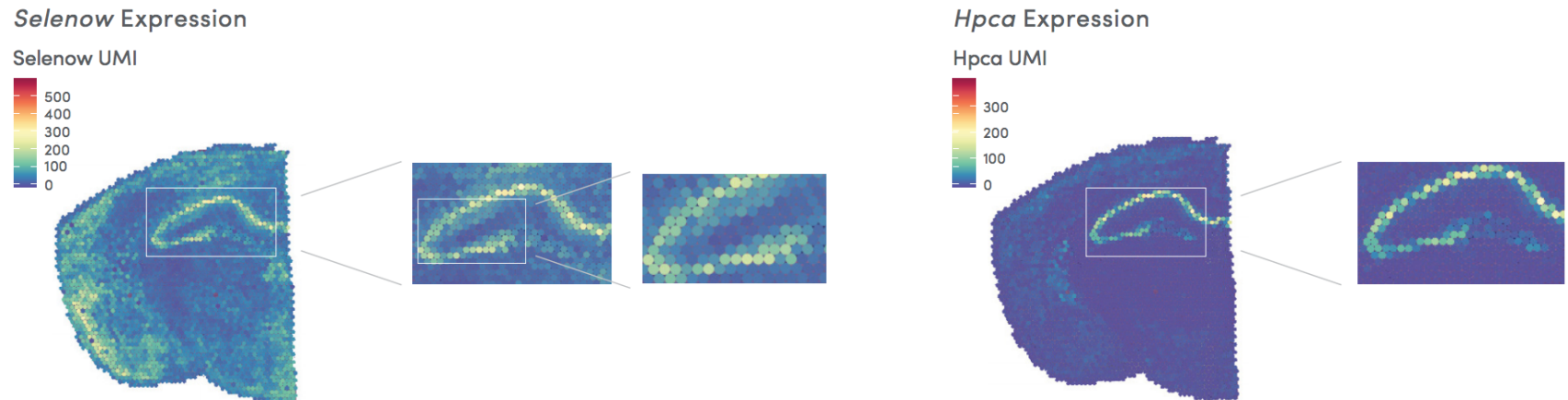


Figure 2. This is a coronal mouse brain section with overlaid spatial gene expression information. The spots correspond to localized mRNA of *Selenow* and *Hpca*, both known to have predominant hippocampal expression.

scSummary

Single cell analysis is a powerful tool to study heterogeneous tissues

- Overcomes fundamental problems that can arise when averaging
- scRNA analysis used to identify novel cell types, understand the progression from one cell type to another across development or disease
- Many other sc-assays in development, expect 1000s to 1Ms of cells in essentially any assay

Major challenges

- Very sparse amplification and few reads per cell
- Find large CNVs, identify major cell types; hard to find small variants or perform differential expression
- Allelic-dropout and unbalanced amplification hides or distorts information
- Use statistical approaches to smooth results based on prior information or other cells from the same cell type
- Need new ways to process and analyze millions of cells at a time